

Monetary Transmission through Shadow Banks

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I find that shadow bank money creation significantly expands during monetary-tightening cycles. This “shadow banking channel” offsets reductions in commercial bank deposits and dampens the impact of monetary policy. Using a structural model of bank competition, I show that the difference in depositor clienteles quantitatively explains banks’ different responses to monetary policy. Facing a more yield-sensitive clientele, shadow banks are more likely to pass through rate hikes to depositors, thereby attracting more deposits when the Federal Reserve raises rates. My results suggest that monetary tightening could unintentionally increase financial fragility by driving deposits into the uninsured shadow banking sector. (*JEL* G23, E52)

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This paper proposes a new transmission channel of monetary policy, the shadow banking channel. Standard theories of monetary transmission predict that high interest rates reduce deposit creation (Bernanke and Blinder 1988; Kashyap and Stein 1995; Drechsler, Savov, and Schnabl 2017). In contrast, I show that such relation is reversed for shadow banks: high interest rates surprisingly expand shadow bank deposits.¹ The shadow banking channel arises from

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¹ Former Federal Reserve Chair Ben Bernanke provided a definition of the shadow banking in April 2012: “Shadow banking, as usually defined, comprises a diverse set of institutions and markets that, collectively, carry out

the competition between shadow and commercial banks in a deposit market with heterogeneous depositors. Facing a more yield-sensitive clientele, shadow banks pass through more rate hikes to depositors, thereby attracting more deposits when the Federal Reserve raises rates.

The shadow banking channel is important for two reasons. First, in recent years, more than 30% of deposits in the United States are created by shadow banks. The rising importance of shadow banks has raised concerns of policy makers on the potential impact on monetary policy.² My estimates show that shadow banks offset one-third of the reduction in commercial bank deposits during monetary tightening cycles, significantly dampening the traditional channels of monetary policy. Second, the shadow banking channel suggests that monetary policy not only affects the *total amount* of bank deposits but also the *relative shares* between the shadow and commercial banking sectors. Because shadow bank deposits are outside of government safety nets, such as deposit insurance and the discount window, shifts in the relative shares of deposits have important implications for financial stability.

I start my analysis by decomposing the aggregate money supply in the United States into a commercial and a shadow banking component. Shadow bank money supply mainly consists of liquid claims created by money market funds (MMFs), which are shadow banks in the deposit market.³ Contrary to the conventional wisdom of commercial banks, I find that shadow bank deposits surprisingly expand when the Fed raises rates. The contrast between shadow and commercial banks can be easily seen in a time-series plot of deposit growth rates in Figure 1.

To understand the underlying mechanism and study policy counterfactuals, I develop a structural model of the shadow banking channel. The prior literature on monetary transmission often assumes banks and depositors are homogeneous. I introduce product differentiation for bank deposits and heterogeneous preferences for depositors following the industrial organization literature (Berry 1994; Berry, Levinsohn, and Pakes 1995; Nevo 2001). In my model, banks are differentiated by their respective degrees of transaction convenience and yields. Shadow banks offer lower transaction convenience compared to commercial banks because the lack of bank charters prohibits them from operating branch networks and payment systems. Instead, they

traditional banking functions but do so outside, or in ways only loosely linked to, the traditional system of regulated depository institutions.”

² For instance, in July 2014, the then-Federal Reserve Board Chair, Janet Yellen said, “We won’t be able to detect them (shadow banks), and if we can, we won’t have adequate regulatory tools. That is a huge challenge to which I don’t have a great answer.”

³ In this paper, I use “MZM” (money zero maturity) as the measure of money supply in the economy. This measure is a modification of M2 introduced after the usefulness of previous measures became compromised in the 1990s. This measure includes currency, traveler’s checks of nonbank issuers, demand deposits, other checkable deposits, savings deposits, retail MMF shares, and institutional MMF shares. However, choosing a specific definition of money aggregates is not important, because my research question concerns each component of the money aggregates rather than the sum.

differentiate themselves by competing on yields. Product differentiation between shadow and commercial banks results in different clienteles for each banking sector. Commercial banks attract a group of transaction-oriented depositors who value transaction services but are insensitive to yields. Typical examples of transaction-oriented depositors include small and unsophisticated depositors who choose banks mainly based on geographical proximity rather than yields. In contrast, shadow banks attract a group of yield-oriented depositors, such as wealthy individuals and corporate treasurers. These yield-oriented depositors are not primarily concerned with transaction convenience, but instead, are sensitive to yields.

Depending on their depositor clientele, commercial and shadow banks strategically set their deposit rates to maximize profits. When the Fed funds rates are low, both types of banks offer similar rates, because commercial banks cannot offer rates much lower than zero given the competition from cash, while shadow banks cannot offer rates much higher than zero given the low returns on assets. In this environment, most depositors choose commercial banks. However, when the Fed raises interest rates, deposit rates of the two banking sectors start to diverge. Commercial banks keep paying low deposit rates, because their main clientele, the transaction-oriented depositors, are attached to their transaction services. In contrast, shadow banks raise deposit rates to keep their yield-sensitive depositors from switching to bonds. As a result, monetary tightening widens the spread between shadow and commercial bank deposit rates, inducing some of the depositors from commercial banks to switch to shadow banks.⁴ This gives rise to the shadow banking channel, in which shadow bank deposits expand when the Fed tightens monetary policy.

The key institutional feature that generates the shadow banking channel is the difference in depositor clienteles between shadow and commercial banks. However, many other institutional differences may generate predictions in the same direction, making it challenging to quantify their relative contributions using a reduced-form method. I overcome this challenge by developing a structural model that incorporates key alternative channels of monetary policy, including the classical bank lending channel based on reserve requirements (Bernanke and Blinder 1988; Kashyap and Stein 1995) and the deposits channel based on the market power of commercial banks (Drechsler, Savov, and Schnabl 2017).

I estimate my model using institution-level data on U.S. commercial banks and MMFs. Three main findings stand out: First, the estimates confirm the view that reserve requirements are not quantitatively important anymore after the 1990s (Bernanke and Gertler 1995; Woodford 2010; Drechsler, Savov, and

⁴ Even though commercial banks lose some deposits to shadow banks, commercial banks prefer to not replicate the shadow banks' strategy, because passing through more rates would reduce their profit margin. On the other hand, shadow banks cannot copy the commercial banks' strategy, because shadow banks cannot offer the same transaction services that keep depositors attached to them.

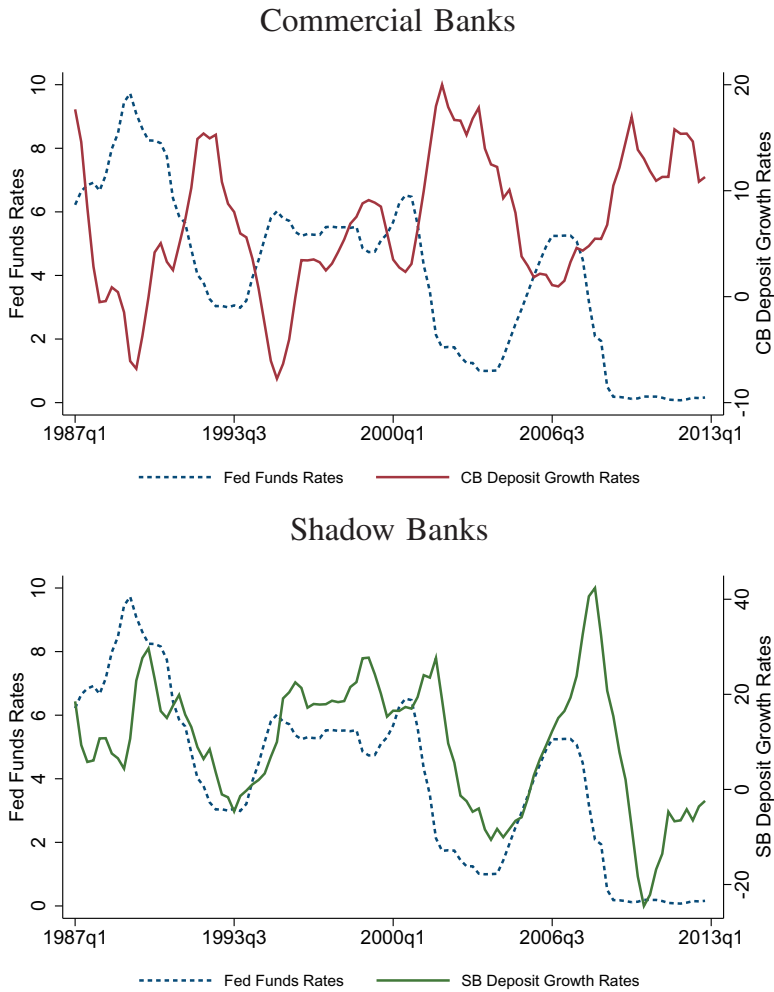


Figure 1
Deposit growth rates and the Fed funds rates
This figure shows the annual growth rates of U.S. commercial and shadow bank deposits from 1987 to 2012. The data are quarterly. Commercial bank deposits are the sum of checking and savings deposits. Shadow bank deposits include all U.S. retail and institutional MMF shares. The data come from FRED.

Schnabl 2017). Instead, most of the impact of monetary policy on commercial banks goes through the deposits channel. Second, the shadow banking channel significantly dampens the deposits channel, offsetting one-third of the reduction of commercial bank deposits during monetary tightening cycles. This result suggests a new explanation for the diminished monetary impact since the 1990s as documented in the macroeconomic literature (Boivin, Kiley, and Mishkin 2011). Third and finally, the main factor that gives rise to different

responses to monetary policy by the two banking sectors is their different depositor clienteles. Differences in reserve requirements or default risks are not quantitatively important.

The structural model allows for a set of policy counterfactuals. I show that an increase in banking concentration, tighter liquidity regulation, or an increase in the fraction of yield-sensitive depositors can increase the impact of monetary policy. These counterfactuals are informative for the ongoing structural changes in the U.S. banking system. I also use the structural model to quantify the impact of shadow banks on depositor surplus. The estimates reveal that the competition in the U.S. commercial banking sector is far from perfect, and the supply of depository services by commercial banks is substantially below the first best case. Against this backdrop, shadow banks significantly increase depositor surplus by making the deposit market more competitive.

This paper first contributes to the literature on banking channels of monetary transmission. So far, this literature has mainly focused on commercial banks (Bernanke and Blinder 1988; Kashyap and Stein 1995, 2000; Drechsler, Savov, and Schnabl 2017). This paper brings shadow banks to the forefront of the theoretical and empirical analysis of monetary policy. I find that shadow banks respond to monetary policy in a way that counteracts the response of commercial banks: they expand rather than contract when the Fed tightens monetary policy. This result is not driven by reserve requirements or deposit rate ceilings on commercial banks as suggested by Tobin and Brainard (1963), because these regulations have been either loosened or abolished since the 1990s. Instead, it arises from the competition between shadow and commercial banks in a deposit market with heterogeneous depositors.

This paper also adds to a new literature which studies the role of imperfect competition in transmitting monetary policy (Duffie and Krishnamurthy 2016; Drechsler, Savov, and Schnabl 2017; Scharfstein and Sunderam 2017). Specifically, Drechsler, Savov, and Schnabl (2017) propose the deposits channel, in which high Fed funds rates allow banks to exercise market power, resulting in a reduction of money creation. They show the existence of the deposits channel using a reduced-form approach. My paper's second contribution is to quantify the magnitude of the deposits channel through a structural model. The estimates show that most of the variation in deposit spreads of commercial banks over monetary cycles comes from the deposits channel.

Note that the deposits channel does not differentiate between shadow and commercial banks. Why shadow banks would not exercise market power in the same way as commercial banks is unclear. The third contribution of this paper is to show that depositor clientele has a large effect on banks' market power. Such effect is not captured by reduced-form measures of market power, such as the Herfindahl-Hirschman index (HHI), but can be measured through a structural IO framework (see Berry, Levinsohn, and Pakes 1995).

This paper also sheds light on an older literature on the classical bank lending channel. [Bernanke and Blinder \(1988\)](#) and [Kashyap and Stein \(1995\)](#) argue that monetary policy influences deposit creation through the reserve requirement on commercial banks.⁵ Although the reserve requirement may have been important historically, many have expressed doubts about its quantitative relevance in recent decades because of technological innovation and regulatory reform after the 1990s ([Bernanke and Gertler 1995](#); [Woodford 2010](#); [Drechsler, Savov, and Schnabl 2017](#)).⁶ My paper contributes to this literature by formally quantifying the effect of the reserve requirement on monetary transmission in the banking system. My estimates show that the effect of reserve requirement is quantitatively small, at least in the post-1990 sample.

This paper also contributes to the literature which studies the interaction between monetary policy and macroprudential policies. After the 2008–2009 financial crisis, many argue that by tightening monetary policy, the central bank can curb the creation of money-like liabilities by the banking system ([Borio and Zhu 2012](#); [Stein 2012](#); [Ajello et al. 2015](#)). My findings show that monetary tightening may unintentionally drive deposits from the insured commercial banking sector to the uninsured shadow banking sector, amplifying the systemic risk. My paper supports the view that “monetary policy is too blunt a tool to address possible financial imbalances” ([Bernanke 2011](#); [Yellen 2014](#)).

Finally, this paper adds to a new and growing body of literature that applies a structural IO approach to study financial intermediation ([Egan, Hortaçsu, and Matvos 2017](#); [Egan, Lewellen, and Sunderam 2017](#); [Koijen and Yogo 2016](#); [Buchak et al. 2017](#)). The structural approach allows quantification of competing channels and analysis of counterfactual policy experiments. This paper is particularly related to [Egan, Hortaçsu, and Matvos \(2017\)](#), who use a similar structural IO framework to study deposit competition and bank runs. In contrast, this paper is the first attempt to use a structural IO model to study transmission channels of monetary policy. [Egan, Hortaçsu, and Matvos \(2017\)](#) assume that different types of deposits (insured versus uninsured) are in separate markets with different depositor clienteles. There is no direct competition among different types of deposits. In contrast, in my model, shadow and commercial deposits compete in the same market. The competition between shadow and commercial banks is important for understanding the transmission of monetary policy.

⁵ The classical bank lending channel makes three key assumptions: (1) the Fed is able to influence deposit creation through the reserve requirement; (2) banks cannot perfectly insulate loan supply from shocks to deposits by borrowing nonreservable wholesale funds because of financing frictions; and (3) bank loans and bonds are imperfect substitutes so the quantity of bank loans matters independent of bond market interest rates.

⁶ The sweep technology developed in the early 1990s allows banks to automatically move funds from checking accounts to saving accounts. This technology reduces required reserves.

1. Deposit Creation by Shadow Banks

In this section, I briefly describe the institutional background of the shadow banking system. I then present several new stylized facts about the shadow banking channel.

1.1 Institutional background

The shadow banking system is a collection of financial intermediaries that conduct maturity, credit, and liquidity transformation outside the traditional commercial banking system. Examples of shadow banks include securitization vehicles, asset-backed commercial paper (ABCP) conduits, MMFs, broker-dealers, and mortgage companies. Like commercial banks, shadow banks transform long-term illiquid assets into short-term money-like claims. Because households and businesses prefer liquidity, issuing money-like claims allows shadow banks to lower their financing costs.

Figure 2 provides a simplified representation of the U.S. banking system.⁷ The upper branch represents the commercial banking sector, whereas the lower branch represents the shadow banking sector. Unlike commercial banks, which combine deposit creation and loan origination under one roof, the shadow banking system separates the intermediation process into different entities. MMFs constitute the first stage of the shadow banking intermediation process. MMFs take deposits from households and businesses and then pass the proceeds to downstream shadow banks, such as securitization vehicles, mortgage conduits, broker-dealers, and mortgage companies that specialize in loan origination.⁸ In this process, MMFs create money-like liabilities—MMF shares—that resemble commercial bank deposits.

MMF shares are widely (though not necessarily accurately) regarded as being as safe as bank deposits while providing a higher yield. Similar to commercial bank deposits, MMFs provide intraday liquidity, and some of them even allow depositors to write checks on their deposits. Because of their similarity to commercial bank deposits, MMF shares are included in official money supply statistics. The amount of MMF shares also provides a good proxy of the quantity of funds flowing into the shadow banking sector. In this paper, I will mainly focus on the competition between MMFs and commercial banks in the deposit market.

While shadow bank deposit creation is conducted by MMFs, loan origination is conducted by different shadow banking entities, such as funding corporations, finance companies, mortgage companies, captive financial institutions, and broker-dealers. For example, Quicken Loans and PHH are shadow banks

⁷ Pozsar et al. (2010) offer a more detailed description of the shadow banking intermediation process.

⁸ Some intermediation chains in the shadow banking system consist of multiple steps. For instance, MMFs could first lend to ABCP conduits, which, in turn, lend to mortgage companies. For the sake of simplicity, Figure 2 combines downstream shadow banks in one block.

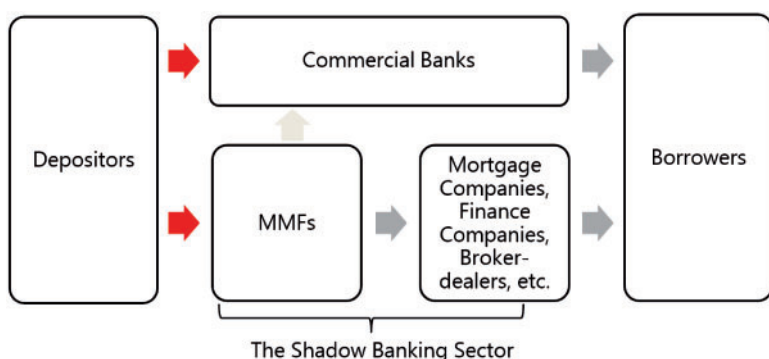


Figure 2
U.S. banking system

that specialize in loan origination in the mortgage market. These downstream shadow banks do not issue deposits directly to depositors. Instead, they obtain a significant amount of funding from MMFs through issuing money market instruments.

Over the past 30 years, the shadow banking sector has become increasingly important in the economy. Based on the aggregate money supply statistics from the Federal Reserve, the share of shadow bank deposits has increased from around 15% in the 1980s to around 40% in 2007, while the share of commercial bank deposits is on a downward trend.

1.2 Data sources

The first main database used in this paper is iMoneyNet. This data set provides monthly share-class level data for U.S. MMFs since 1985. After a cross-check with the aggregate money supply statistics from the Federal Reserve Board, I find that this database covers essentially all the MMFs after 1987. The data contain detailed information on fund characteristics, such as deposit amounts, yields, incurred management costs, and other incurred operating costs.⁹ Portfolio holding information became available in 1998 and includes average portfolio maturity and portfolio weights by asset class. As data on shadow banks are generally very scarce, this data set provides a rare glimpse into the inner workings of the shadow banking system.

The second main data set is the Consolidated Report of Condition and Income, generally referred to as the Call Report. This data set provides quarterly bank-level data for every insured U.S. commercial bank, including detailed accounting information, such as deposit amounts, interest income, salary expenses, and fixed-asset expenses. I complement the Call Report with

⁹ The incurred management costs and other incurred operating costs are not charged to depositors. They are incurred by the MMFs.

the FDIC Summary of Deposits, which provides the information on the number of branches of each bank going back to 1994. Following the literature, I impute deposit rates from bank financial statements by dividing deposit interest expenses over the total amount of deposits (Dick 2008).

In addition to the two main data sources above, I also use the Survey of Consumer Finance (SCF) 2013 to obtain depositor-level deposit holdings and demographic information. Finally, I retrieve the aggregate time series of the amount of cash and total financial wealth held by U.S. households and the Fed funds rates from the Federal Reserve Economic Data (FRED).

I construct two measures of monetary policy shocks. The first one is the raw changes in the Fed funds rates. The second is the Romer and Romer (2004) exogenous shocks to the Fed funds rates.¹⁰ Romer and Romer (2004) use quantitative and narrative records to infer the Federal Reserve's intentions for the Fed funds rate around FOMC meetings. The series is constructed by purging out endogenous and anticipatory movements from the raw changes in the Fed funds rates, which helps to alleviate concerns on the endogeneity of monetary policy. The original Romer and Romer (2004) series ends in 1996. I extend the series to 2012 using the subsequent Greenbook data.

The final sample for structural estimation is from 1995 to 2012. The unit of observation is bank-year. I construct market shares of each bank in each year in the national market. The total market size is total financial wealth of U.S. households in a given year. The sample construction consists of two steps. In the first step, I combine banks with market shares less than 0.001% with the outside option.¹¹ In the second step, I follow Egan, Hortaçsu, and Matvos (2017) to drop banks with fewer than ten domestic branches to filter out foreign banks in the remaining sample.¹² In addition to deposit rates and market share, I also create a set of bank characteristics, such as number of branches, number of employees per branch, expenses of fixed assets, salary expense, reserves, check-writing dummy, and a stand-alone fund dummy, incurred management costs, and other incurred operating costs. Note that different types of deposits face different reserve requirements. I calculate the weighted average reserves ratio for each bank using the actual amount of reserves divided by the amount of deposits.

Table 1 provides the summary statistics of the final sample used for the structural estimation. Commercial banks on average offer lower deposit rates than shadow banks: the average deposit rates are 1.72% for commercial banks and 2.85% for MMFs. Commercial banks in the sample on average have 106 branches nationwide. Each branch on average has 21 employees. Commercial banks on average incur fixed asset expenses of 0.4% of total assets and salary

¹⁰ Thanks to the anonymous referee for making this suggestion.

¹¹ Nevo (2001) shows that the estimates of price sensitivity for the inside goods are not biased by the choice of outside goods.

¹² The second step excludes around 70 banks per year, which account for 7% of the deposits.

Table 1
Summary statistics

	Mean	SD	p25	p50	p75
Cash					
Amount	657.620	203.140	485.754	664.021	776.142
Market share	1.601	0.119	1.486	1.599	1.696
Commercial banks					
Amount	10.308	60.599	0.755	1.332	3.341
Market share	0.024	0.123	0.002	0.003	0.008
Deposit rates	1.723	1.166	0.737	1.710	2.683
Number of branches	109.826	385.415	18.000	28.000	61.000
Employees per branch	20.418	19.621	12.503	16.000	21.260
Expenses of fixed assets	0.454	0.149	0.341	0.437	0.558
Salaries	1.611	0.444	1.270	1.559	1.897
Reserves	1.871	4.039	0.135	0.492	1.594
Money market funds					
Amount	6.228	10.388	1.189	2.644	6.832
Market share	0.015	0.023	0.003	0.007	0.017
Deposit rates	2.807	2.003	0.798	2.932	4.778
Check-writing dummy	0.463	0.499	0.000	0.000	1.000
Standalone dummy	0.202	0.402	0.000	0.000	0.000
Incurred management costs	0.259	0.161	0.147	0.250	0.395
Other incurred operating costs	0.161	0.150	0.053	0.120	0.223

This table presents summary statistics of a sample of commercial banks and MMFs from 1995 to 2012 in the United States. Expenses of fixed assets and salaries are normalized by total assets. The total market size is defined as the aggregate financial assets of households. Deposit amount is in billions of dollars. Deposit rates, market shares, expenses of fixed assets, salaries, reserves, management costs, and other costs are expressed as percentages.

expense of 1.6% of total assets. The average effective reserves ratio is 1.9%. Of the sample MMFs, 46% offer some form of check-writing services.¹³ MMFs on average incur management costs of 0.26% of total assets and other operating costs of 0.16% of total assets.

1.3 Stylized facts

In this section, I document a set of new stylized facts on monetary transmission through the shadow banking system. Following the intermediation chain in Figure 2, I first study how monetary policy affects the shadow bank money creation by MMFs (the first arrow). Then I study how MMFs pass the raised deposits to downstream shadow banks (the second arrow). Finally, I study how the lending of downstream shadow banks changes in response to changes in the funding supply (the third arrow).

First, I investigate the effect of monetary policy on shadow bank money creation. Specifically, I break down the aggregate money supply into cash, commercial bank deposits, and shadow bank deposits. Commercial bank deposits include demand and savings deposits. Shadow banking deposits

¹³ The check-writing services offered by MMFs are often limited, because MMFs do not have access to the Master Account with a Federal Reserve Bank to settle transactions, which requires a formal bank charter. The common restrictions are a \$250 minimum per check and 6 checks in total per month.

include retail MMF shares and institutional MMF shares. Figure 1 plots the Fed funds rates and the annual deposit growth rates of each banking sector over time. Conventional monetary transmission channels predict that high Fed funds rates have tightening effects on the money supply (Bernanke and Blinder 1988; Kashyap and Stein 1995, 2000; Drechsler, Savov, and Schnabl 2017). This prediction has been verified by prior literature, which I replicate here. The top panel of Figure 1 shows the growth rates of commercial bank deposit rates and the Fed funds rates. Consistent with conventional wisdom, high Fed funds rates are associated with low growth rates of commercial bank deposits.

What remains unknown is what happens to shadow bank deposit creation. The bottom panel of Figure 1 plots the deposit growth rates of shadow banks. In contrast to conventional wisdom in the commercial banking sector, high Fed funds rates are associated with high growth rates of shadow bank deposits. This means that monetary tightening has an expansionary effect on shadow bank money creation.

Formally, I regress deposit growth rates of each banking sector on two measures of monetary policy shocks: one is the change in the Fed funds rates; the other is exogenous shocks to the Fed funds rates constructed by Romer and Romer (2004). In the baseline regressions, I consider a 3-year horizon in the construction of the change in the Fed funds rates or the Romer and Romer (2004) shocks, because deposit flows respond to monetary policy in a persistent manner.¹⁴ The results are robust to alternative horizons. I control for a list of macroeconomic variables, such as gross domestic product (GDP) growth rates, inflation, the TED spread, growth rates of corporate cash holdings, and a time trend. I also include two measures of regulatory tightness: a dummy variable for the Gramm-Leach-Bliley Act, which loosens financial regulation; a dummy variable for the Dodd-Frank Act, which tightens financial regulation. The regression model is the following:

$$\text{Deposit Growth Rates}_t = \alpha + \beta \text{Monetary Policy Shocks}_t + \gamma X_t + \epsilon_t. \quad (1)$$

Table 2 reports the regression results. Columns 1–4 use the changes in the Fed funds rates as monetary policy shocks, and Columns 5–8 use the Romer and Romer (2004) exogenous monetary shocks. Consistent with the graphical observation, monetary policy has opposite effects on these two sectors: an increase in the Fed funds rates is associated with a decrease in the growth rates of commercial bank deposits, but an increase in the growth rates of shadow bank deposits. The effect of monetary policy on total money supply is still negative, but the magnitude is quite small, because shadow bank deposit creation partially offsets the reduction of commercial bank deposits and attenuates the impact of monetary tightening on aggregate money supply. The result is robust to using

¹⁴ Online Appendix Figure 1 shows that a shock to the Fed funds rates in year 0 leads to a persistent increase in shadow bank deposits that peaks around year 3.

either the raw changes in the Fed funds rates or the [Romer and Romer \(2004\)](#) shocks.¹⁵

To understand why monetary tightening has an expansionary effect on shadow bank deposits, I examine deposit rates paid by commercial and shadow banks in the upper panel of Figure 3. To address the concern that differences in rates paid by shadow and commercial banks reflect differences in risk, I adjust shadow bank deposit rates by subtracting a composite of credit spreads of money market instruments weighted by their portfolio weights in MMFs. I adjust commercial bank deposit rates by subtracting the TED spreads multiplied by the share of uninsured deposits.¹⁶ I find that the spread between shadow and commercial bank deposit rates widens when the Fed increases interest rates. The effect is economically significant. For example, in the 2004–2006 tightening cycle, the spread between shadow and commercial bank deposit rates increased from less than 0.5% to nearly 3%. As transaction convenience of bank deposits is relatively stable over monetary cycles, such a big increase in the relative yields may make shadow banks more attractive. This seems to explain the expansion of shadow banks in periods of high interest rates.

It is interesting to contrast the behavior of deposit rates to lending rates. The lower panel of Figure 3 plots the average lending rates of 30-year fixed-rate mortgages issued by commercial banks and shadow banks. Shadow banks considered here are mortgage companies, such as Quicken Loans and PHH. Unlike deposit rates, which significantly diverge over monetary policy cycles, lending rates seem to move in tandem over monetary cycles across the two types of banks.¹⁷ The comparison between deposit and loan rates suggests that the different responses to monetary policy of the two types of banks seem to originate from the deposit market.

So far I have shown that monetary tightening seems to expand money creation by shadow banks. Now I study the impact of monetary policy on the lending of MMFs to downstream shadow banks (the second arrow in Figure 2). MMFs lend to downstream shadow banks through money market instruments, such as commercial paper (CP), asset-backed commercial paper (ABCP), repurchase agreements (repo), and short-term notes. Figure 4 shows breakdown of MMF lending by type of security. The first major component is commercial

¹⁵ One may worry that unconventional monetary policy post-2008 could drive the result, so I repeat the analysis using the pre-2008 sample. As shown in [Online Appendix Table 1](#), the result is very similar to the full sample analysis. One may also wonder whether the result may differ across retail MMFs and institutional MMFs. Figure 2 of the [Online Appendix](#) plots the deposit growth rates of retail and institutional MMFs separately. The cyclical pattern is quite similar for both retail and institutional MMFs, whereas the magnitude is larger for institutional ones.

¹⁶ Note that repo, ABCP, and commercial paper rates are not available for the full 1987–2012 sample period. I calculate the average spread of each of these rates over 3-month LIBOR in the sample from 2005 to 2012 and extrapolate to earlier periods assuming the spreads relative to LIBOR are constant. The sample average of the share of uninsured deposits is 0.45 based on the FDIC Statistics on Depository Institutions. The [Online Appendix](#) shows the results are similar when unadjusted deposit rates are used.

¹⁷ Lending rates of other types of loans also move in a similar way over the monetary cycles, as reported in [Online Appendix Figure 3](#).

Table 2
Monetary policy and money growth

	(1) CB	(2) MMF	(3) Cash	(4) Total	(5) CB	(6) MMF	(7) Cash	(8) Total
Monetary policy shocks	-1.417*** (0.147)	1.986*** (0.375)	-0.114 (0.089)	-0.326* (0.184)	-2.521*** (0.320)	3.059*** (0.789)	-0.390** (0.173)	-0.655* (0.364)
GDP growth	-0.065 (0.247)	-1.813*** (0.630)	-0.242 (0.150)	-0.488 (0.309)	-0.777*** (0.263)	-0.834 (0.649)	-0.307** (0.142)	-0.655** (0.299)
Inflation	-0.068 (0.395)	-1.104 (1.009)	-0.992*** (0.240)	-0.529 (0.495)	-0.319 (0.430)	-0.515 (1.061)	-0.919*** (0.232)	-0.549 (0.489)
TED spread	-1.991 (1.198)	17.736*** (3.061)	0.390 (0.728)	5.155*** (1.502)	-3.857*** (1.291)	20.469*** (3.187)	0.287 (0.698)	4.744*** (1.468)
Corporate cash	0.059 (0.058)	0.237 (0.148)	-0.034 (0.035)	0.098 (0.073)	0.097 (0.064)	0.193 (0.158)	-0.028 (0.035)	0.108 (0.073)
Gramm-Leach-Bliley	3.218* (1.729)	2.038 (4.417)	1.449 (1.050)	3.507 (2.167)	-0.059 (2.044)	5.395 (5.046)	0.697 (1.105)	2.557 (3.325)
Dodd-Frank	2.599 (1.703)	-4.811 (4.351)	4.845*** (1.035)	1.708 (2.135)	8.075*** (1.881)	-11.969** (4.645)	5.488*** (1.017)	3.049 (2.140)
Observations	92	92	92	92	92	92	92	92
Adj. R^2	.693	.624	.466	.158	.628	.574	.487	.159

This table presents time-series regressions of aggregate annual money growth rates on monetary policy shocks. Monetary policy shocks are measured as 3-year cumulative changes in the Fed funds rates in Columns 1–4 and 3-year cumulative [Romer and Romer \(2004\)](#) shocks in Columns 5–8. A time trend is also included in the regressions. The data frequency is quarterly. The sample period is from 1990 to 2012. Standard errors are given in parentheses and are computed with Newey-West standard errors with 12 lags. * $p > .1$; ** $p > .05$; *** $p > .01$.

paper and ABCP (22%). ABCP conduits and special purpose vehicles issue these instruments to finance residential mortgages and asset-backed securities originated by mortgage companies and banks ([Acharya, Schnabl, and Suarez, 2013](#)).¹⁸ In aggregate, MMFs account for 37% of all outstanding commercial paper and 26% of all outstanding ABCP in the 1998–2012 sample period.¹⁹ The second major component of MMFs' portfolio is repurchase agreement (16%). These instruments are mainly used by broker-dealers to finance their lending to hedge funds and special purpose vehicles. As a whole, MMFs account for around 50% of the repo lending to downstream shadow banks.²⁰ The third component is short-term notes (24%), which are usually used by captive finance companies within large corporations. Adding up the above three components, 62% of MMFs' portfolio can be attributed to lending to downstream shadow

¹⁸ The data on downstream shadow banks are sparse, because many downstream shadow banks are private entities with little public information. Nevertheless, the SEC filings of a publicly listed mortgage company, PHH Corporation, reveal that this mortgage company heavily relied on money market instruments, such as repo and ABCP, to finance its lending (see Form 8-K 2006, Form 10-K 2007, PHH Corporation). Note that commercial banks also sell some of their mortgages to ABCP conduits. To the extent that these mortgages are financed by uninsured shadow bank deposits, they also can be interpreted as a type of shadow banking activity.

¹⁹ Moody's Investors Service Special report on July 8, 2002, notes that "as of May 31, U.S. money market funds held more than \$343 billion of ABCP, or over 48% of all outstanding ABCP." As a result, U.S. money market funds "carry a big 'buy-side' stick with ABCP."

²⁰ See table 1 of [Krishnamurthy, Nagel, and Orlow \(2014\)](#). Note this calculation does not include interbank repo, because, as argued by [Krishnamurthy, Nagel, and Orlow \(2014\)](#), "the amount of interbank repo is informative about the length of intermediation chains, but not about the amount of credit received from and extended to the rest of the economy."

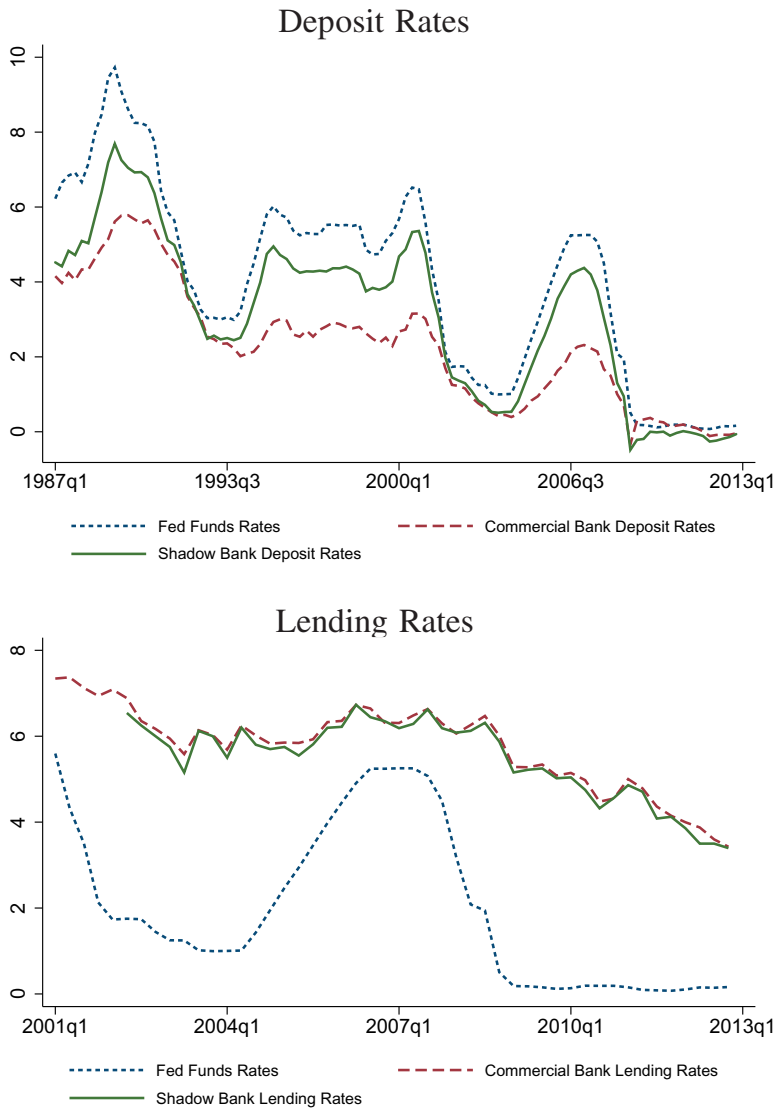


Figure 3
Deposit and lending rates of commercial and shadow banks

The upper panel shows the average deposit rates of U.S. commercial banks and shadow banks. Commercial bank deposit rates are the average interest returns of demand and saving deposits. Shadow bank deposit rates are net yields of MMFs. The lower panel shows the interest rates of 30-year fixed rate mortgages originated by commercial banks and shadow banks (mortgage companies). The data are quarterly. The data come from the Call Report, iMoneyNet, and RateWatch.

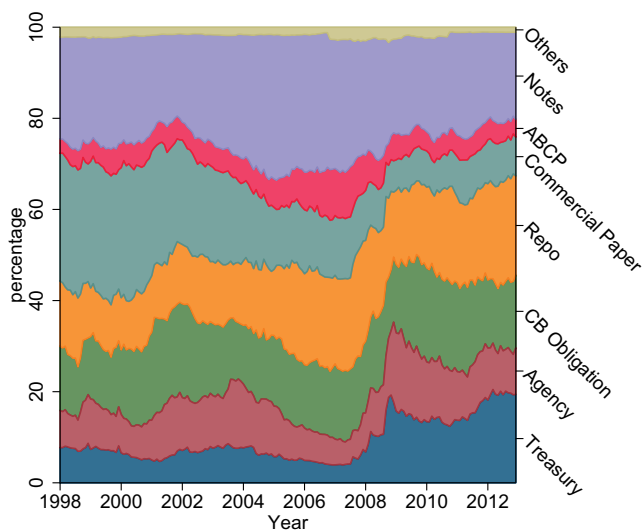


Figure 4

Composition of MMFs' lending portfolio over time

The sample period is from 1998 to 2012. The data are monthly. The data come from the iMoneyNet.

banks. Besides lending to shadow banks, 16% of MMFs' funding circles back to commercial banks through large denomination bank CDs; 9% goes to U.S. Treasuries; and 10% goes to debt issued by government agencies, such as Fannie Mae and Freddie Mac.

I conjecture that as more deposits flow into MMFs, lending from MMFs to downstream shadow banks should increase. To verify this conjecture, I regress annual growth rates of MMF lending by each type of money market instruments on monetary policy shocks measured by [Romer and Romer \(2004\)](#) series controlling for macroeconomic variables, fund characteristics, and fund fixed effects. The macroeconomic controls include GDP growth rates, inflation, TED spread, growth rates of corporate cash holdings, and measures of regulatory tightness. The fund characteristics include fund size, fund age, management costs, and other costs.

$$MMF \text{ Lending Growth Rates}_{i,t} = \alpha + \beta \text{Monetary Policy Shocks}_t + \gamma X_{i,t} + \epsilon_{i,t}. \quad (2)$$

Columns 1 to 4 of Table 3 show that MMFs significantly increase their lending to downstream shadow banks as the Fed funds rate increases. The economic magnitude is significant: a 1% increase in the Fed funds rates is associated with a 0.20%–0.91% increase in lending from MMFs to other shadow banks. In addition to the four types of money market instruments discussed above, MMFs also hold commercial bank obligations, which are issued by commercial banks to obtain short-term funding. Column 6 of Table 3

shows that MMFs also increase the holding of large denomination bank obligations when the Fed raises interest rates. This result reveals an interesting interaction between the shadow and commercial banking system. As the Fed tightens monetary policy, commercial banks borrow more from MMFs to compensate for their loss of core deposits.²¹ This may have implications for financial stability as core deposits are often insured but large denomination bank obligations are not.

Last, I examine the final step of the intermediation chain—the lending of downstream shadow banks to ultimate borrowers. I conjecture that the downstream shadow banks should be able to expand their credit supply with an increase in funding supply from MMFs. To verify this conjecture, I regress the asset growth rates of five types of shadow banks, that is, funding corporations, finance companies, ABCP issuers, captive financial institutions, and broker-dealers on MMF deposit growth rates instrumented by [Romer and Romer \(2004\)](#) monetary policy shocks. Control variables similar to Equation (1) are included.

$$\text{Shadow Bank Asset Growth}_t = \alpha + \beta \text{MMF Deposit Growth}_t + \gamma X_t + \epsilon_t. \quad (3)$$

Table 4 presents the results. I find that an increase in the MMF deposit growth indeed leads to an expansion of lending in downstream shadow banks. A 1% increase in the MMF deposit growth rate is associated with a 0.54% increase in the asset growth rate of the downstream shadow banks combined.

To summarize, this section finds that shadow bank deposits expand during periods of monetary tightening while commercial bank deposits contract. The different responses in the quantity of deposits seem to be related to the spread between shadow and commercial bank deposit rates, which usually widens when the Fed raises interest rates.

2. A Structural Model of Bank Competition

2.1 Intuition

To rationalize the above empirical findings, I introduce imperfect competition between differentiated banks following [Berry, Levinsohn, and Pakes \(1995\)](#) (BLP) into a model of banking monetary transmission as formulated by [Bernanke and Blinder \(1988\)](#). The model has two key ingredients. First, commercial and shadow bank deposits offer different degrees of transaction convenience. Specifically, commercial bank deposits provide greater convenience because of their branch networks, ATMs, and payment systems.²² In contrast, shadow bank deposits offer less convenience because

²¹ [Choi and Choi \(2016\)](#) examine the balance sheets of commercial banks and reach a similar conclusion.

²² In addition, the deposit insurance on commercial bank deposits may also increase their convenience relative to shadow banks. Deposit insurance is less relevant for very large depositors, because the FDIC only insures commercial bank deposits up to a certain amount.

Table 3
Monetary policy and MMF lending

	(1) Commercial paper	(2) ABCP	(3) Repo	(4) FRNs	(5) Treasury & agency	(6) Bank obligations
Monetary policy shocks	0.914*** [0.121]	0.313*** [0.049]	0.783*** [0.114]	0.201** [0.089]	-0.451*** [0.117]	0.683*** [0.095]
GDP growth	-0.394*** [0.100]	0.185*** [0.038]	0.152 [0.100]	0.331*** [0.088]	-1.308*** [0.136]	-0.394*** [0.091]
Inflation	0.211 [0.158]	-0.211*** [0.068]	0.170 [0.186]	0.053 [0.156]	1.090*** [0.252]	0.348*** [0.155]
TED spread	0.642* [0.387]	0.105 [0.172]	-0.132 [0.471]	0.345 [0.352]	5.552*** [0.588]	1.003*** [0.371]
Corporate cash	0.198*** [0.029]	-0.013 [0.012]	0.078*** [0.029]	-0.044* [0.025]	-0.191*** [0.035]	0.151*** [0.026]
Gramm-Leach-Bliley	-7.734*** [0.698]	-0.847** [0.359]	0.764 [0.572]	-1.694*** [0.550]	-4.387*** [0.735]	-2.866*** [0.704]
Dodd-Frank	-0.469 [0.476]	-1.376*** [0.198]	-3.603*** [0.513]	-3.349*** [0.418]	-2.960*** [0.584]	-3.627*** [0.414]
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	15,060	15,060	15,060	15,060	15,060	15,060
Adj. R^2	.085	.068	.101	.076	.227	.140

This table presents regressions of MMF lending on monetary policy shocks. The dependent variable is the annual change in a specific type of lending normalized by the lagged total lending (lagged 1 year). Monetary policy shocks are measured as 3-year cumulative [Romer and Romer \(2004\)](#) shocks. Fund characteristics include fund size (log), fund age, management costs, and other costs. The sample includes 1,148 MMFs in the period from 1998 to 2012. The data frequency is quarterly. The regressions are weighted by fund assets. Standard errors are given in brackets and are clustered by time. * $p > .1$; ** $p > .05$; *** $p > .01$.

Table 4
Monetary policy and asset growth of shadow banks

	(1) Funding corporations	(2) Finance companies	(3) ABCP issuers	(4) Captive financials	(5) Broker dealers	(6) Shadow bank total
MMF deposits growth	0.468** (0.235)	-0.170 (0.286)	0.715** (0.332)	0.277* (0.141)	0.692 (0.450)	0.540*** (0.182)
GDP growth	3.394*** (0.757)	2.034*** (0.716)	3.140*** (1.068)	1.402** (0.615)	2.352* (1.294)	2.694*** (0.580)
Inflation	-2.034* (1.146)	1.104* (0.617)	1.831 (1.607)	-3.848*** (0.798)	2.784* (1.647)	-0.147 (0.826)
TED spread	11.49** (5.661)	3.829 (6.053)	-19.29** (7.819)	6.029** (2.721)	-25.30** (10.10)	-9.311** (4.589)
Corporate cash	0.0806 (0.194)	-0.0619 (0.170)	-0.467* (0.261)	-0.0236 (0.109)	0.314 (0.273)	-0.0464 (0.144)
Gramm-Leach-Bliley	-3.072 (2.505)	2.751 (3.234)	-3.608 (3.762)	0.743 (1.612)	3.453 (5.096)	2.146 (2.060)
Dodd-Frank	-5.641 (4.139)	-12.35*** (4.011)	-21.54*** (5.025)	-2.206 (3.178)	2.012 (6.223)	-4.899 (3.155)
Observations	92	92	92	92	92	92
Adj. R^2	.681	.190	.621	.547	.235	.552

This table presents time-series regressions of the aggregate asset growth rates of shadow banks on MMF deposit growth rates instrumented by monetary policy shocks. The dependent variable is the asset growth rates of different types of shadow banks. Monetary policy shocks are measured as 3-year cumulative [Romer and Romer \(2004\)](#) shocks. The data frequency is quarterly. The sample period is from 1990 to 2012. Standard errors in brackets are computed with Newey-West standard error with 12 lags. * $p > .1$; ** $p > .05$; *** $p > .01$.

they cannot provide some transaction services due to charter restrictions.²³ To compensate for the lack of transaction convenience, shadow banks usually compete on yields.

The second key ingredient of the model is that depositors exhibit heterogeneous preferences over convenience and yields. There is a group of “transaction-oriented” depositors who care a lot about transaction convenience but who are not very sensitive to yields. For example, “mom and pop” depositors are typical transaction-oriented depositors, who choose banks mainly based on geographical proximity rather than deposit rates paid by banks. There is also a group of “yield-oriented depositors” who are very sensitive to yields but who are relatively insensitive to convenience. For example, large corporations and wealthy individuals are typical yield-oriented depositors. They are less concerned about transaction convenience but are very sensitive to yields.

These two groups of depositors are likely to self-select into different types of banks. Commercial banks are likely to attract more transaction-oriented depositors because of their superior transaction services, while shadow banks attract more yield-oriented depositors because of high deposit rates. Consistent with this idea, using the Survey of Consumer Finances (SCF) 2013, I find that depositors who are wealthy or more sophisticated (proxied by college education) are more likely to choose shadow banks. Table 5 reports the result.

Facing different depositor clienteles, shadow and commercial bank deposit rates exhibit different sensitivities to monetary policy. When the Fed increases interest rates, commercial banks are not pressured to increase their deposit rates because their main depositor clientele—transaction-oriented depositors—are attached to their transaction services. This allows commercial banks to keep deposit rates relatively low and earn higher spreads between lending rates and deposit rates. In contrast, shadow banks have to raise their deposit rates with the market interest rates. Otherwise, their yield-oriented clientele will switch to other higher-yielding liquid assets, such as short-term bonds. As a result, when the Fed raises interest rates, the gap between shadow and commercial bank deposit rates widens, and some of the marginal depositors will switch from commercial banks to shadow banks. This explains why shadow bank deposits expand while commercial bank deposits shrink during monetary tightening. In the following analysis, I refer to this channel as the shadow banking channel of monetary policy.

2.2 Model setting

Having shown the basic intuition of the shadow banking channel, I now describe the model setting.

²³ In practice, some MMFs provide check-writing services by working with commercial banks. However, there are restrictions on the minimum dollar amount for each check and the number of checks allowed per month.

Table 5
Demographic determinants of shadow bank deposit holdings

	(1) Shadow deposit dummy	(2) Shadow deposit share
Income	0.036*** (0.002)	0.020*** (0.001)
College	0.040*** (0.004)	0.014*** (0.003)
Unemployed	0.005 (0.008)	0.002 (0.005)
Age	−0.011*** (0.002)	−0.006*** (0.001)
Age ²	0.001 (0.002)	0.004*** (0.001)
Home owner	−0.010** (0.004)	−0.007** (0.003)
Car owner	−0.010** (0.004)	−0.007*** (0.003)
Female	0.023*** (0.005)	0.018*** (0.004)
Married	0.018*** (0.005)	0.009*** (0.003)
Observations	27764	27764
Adj. <i>R</i> ²	.047	.025

This table presents cross-sectional regressions of shadow bank deposit holdings on demographic variables for a cross-section of 27,764 households in the Survey of Consumer Finance (2013). Shadow bank deposits are defined as deposits that are not insured by the government. Shadow dummy equals 1 if a household has shadow bank deposits and 0 otherwise. Shadow share is the share of shadow bank deposits in the total deposits of a household. The independent variables are the demographics of the head of the household. Robust standard errors are given in parentheses. **p* > .1; ***p* > .05; ****p* > .01.

2.3 Depositors

There are a group of depositors with a measure of 1. Each of them is endowed with one dollar. In each year *t*, depositors make a discrete choice among options including bonds, cash, commercial bank deposits, or shadow bank deposits. Each option *j* is characterized by a deposit rate and convenience pair, (r_j, x_j) .²⁴ Given that shadow banks do not have bank charters that allow them to offer many transaction services, the convenience of shadow bank deposits is lower than commercial bank deposits.

In addition to bank deposits, depositors can also choose cash or bonds. Cash has the highest convenience but zero returns, while bonds have the highest return but no transaction convenience. The return of bonds equals the Fed funds rates, *f*, which are determined by monetary policy.

The optimization problem of the depositor is to choose the option that gives rise to the highest utility. The assumption that each depositor has only one dollar and can choose only one option is not as restrictive as it may appear. We can think as if depositors make multiple discrete choices for each dollar that they have. The probability of choosing each of the options can be interpreted as the portfolio weight. Formally, define the utility of product *j* of depositor *i*

²⁴ Here, I suppress the subscript *t* for simplicity.

as $u_{i,j}$, the depositor's problem is

$$\max_{j \in \{0,1,\dots,J+1\}} u_{i,j} = \alpha_i r_j + \beta_i x_j + \xi_j + \epsilon_{i,j}, \quad (4)$$

where r_j is the deposit rate, x_j is the transaction convenience, ξ_j is the unobservable demand shock for product j , $\epsilon_{i,j}$ is a mean-zero idiosyncratic utility shock for depositor i if choosing product j , which follows the extreme value distribution with a cumulative distribution function $F(\epsilon) = \exp\{-\exp(-\epsilon)\}$. This distribution assumption is standard in structural IO literature. It allows closed-form solution of the choice probabilities. $\{0,1,\dots,J,J+1\}$ is the choice set, where 0 represents cash, $1,\dots,J$ represent commercial banks and shadow banks, and $J+1$ represents bonds. Finally, α_i and β_i are the sensitivity to deposit rates and transaction convenience for each respective depositor type i .

Define $s_{i,j}$ as the choice probability for depositor type i to choose product j . Using the property of the extreme value distribution, the choice probability is given by the following formula:

$$s_{i,j}(r_j|f) = \frac{\exp(\alpha_i r_j + \beta_i x_j + \xi_j)}{\exp(\alpha_i f + \xi_{J+1}) + \exp(\beta_i x_0 + \xi_0) + \sum_{l=1}^J \exp(\alpha_i r_l + \beta_i x_l + \xi_l)}. \quad (5)$$

The numerator is the exponential utility from holding deposits of bank j . The denominator is the sum of exponential utility from all competing products in the market. Intuitively, if bank j generates a higher utility for depositor i (larger numerator), it is more likely to be chosen; if the competitors of bank j generates a higher utility (larger denominator), then bank j is less likely to be chosen. Note that the first term in the denominator, $\exp(\alpha_i f + \xi_{J+1})$, represents the exponential utility from bonds. The second term, $\exp(\beta_i x_0 + \xi_0)$, is the utility of holding cash.

The demand for deposits of bank j is given by summing the choice probability over different depositor types

$$d_j(r_j|f) = \sum_i \mu_i s_{i,j}(r_j|f), \quad (6)$$

where μ_i is the frequency of type i depositors. I normalize the total wealth to \$1, so the demand is the same as the market share. Different types of banks have different demand functions because product differentiation leads to different depositor clienteles.

2.4 Banks

In addition to different demand, commercial and shadow banks have different cost structure. First, commercial banks face a reserve requirement mandates that a fraction λ_j of deposits needs to be held as non-interest-bearing reserves. In contrast, shadow banks face no reserve requirement, so λ_j is zero for shadow

banks. Second, shadow banks incur a lower operating cost, κ_j , than commercial banks because they do not operate branch networks.

Given the demand and cost structure, banks choose deposit rates to maximize their profits.²⁵

$$\max_{r_j} ((1 - \lambda_j)(f + l_j) - r_j - \kappa_j) d_j(r_j | f), \quad (7)$$

where λ_j is the reserve ratio, f is the Fed funds rate, l_j is the lending spread, $(1 - \lambda_j)(f + l_j)$ is return on assets for bank j , κ_j is the marginal operating cost, r_j is the deposit rate of bank j , and $d_j(r_j | f)$ is the demand function for bank j 's deposits.²⁶ The first-order condition gives rise to the classical pricing equation in the IO literature:

$$\begin{aligned} \text{FOC:} \quad & \underbrace{f - r_j}_{\text{Deposit spreads } (p_j)} = \underbrace{\kappa_j + \lambda_j f - (1 - \lambda_j)l_j}_{\text{Marginal costs } (c_j)} + \underbrace{\left(-\frac{\partial \log(d_j(p_j | f))}{\partial p_j} \right)^{-1}}_{\text{Markup } (m_j)} \end{aligned} \quad (8)$$

The left hand side is the deposit spreads, p_j , which is the price that depositors pay for the depository services. The deposit spreads can be decomposed into a marginal cost term and a markup term. The first term on the right-hand side is the marginal cost of providing depository services, c_j . It equals the sum of the operating cost, κ_j , and the opportunity cost of holding reserves, $\lambda_j f$, net the lending spread adjusted by the reserve ratio, $(1 - \lambda_j)l_j$.²⁷ The second term on the right-hand side, $\left(-\frac{\partial \log(d_j)}{\partial p_j} \right)^{-1}$, is the markup that a bank charges on its depository service over the marginal cost. It is inversely related to the demand elasticity. If the demand is inelastic, then the bank can charge a higher markup.

To close the model, the lending spreads are determined by the supply and demand for loans:

$$L_j(l_j) = (1 - \lambda_j)d_j, \quad (9)$$

where l_j is the lending spread of bank j , $L_j(\cdot)$ is the demand function for the loan of bank j with $L' < 0$, λ is the reserve ratio for bank j , and $(1 - \lambda_j)d_j$ is the loan supply.

²⁵ As discussed in Section 1, in practice, MMFs do not directly lend to ultimate borrowers. Instead, they pass deposits to downstream shadow banks that originate loans. Here, the formulation considers the whole of shadow banking.

²⁶ Here, an implicit assumption is that banks cannot conduct perfect price discrimination, that is, offering a different deposit rate to each type of depositor. In reality, banks do try to conduct some form of price discrimination by setting up different types of accounts. However, one can reasonably believe such price discrimination is far from perfect, because banks still face a trade-off between deposit volumes and rates. Therefore, the deposit rate here should be thought of as the weighted average deposit rates of all types of accounts of a bank.

²⁷ After October 2008, the Fed started to pay interest on reserves, r_{IOR} . The opportunity cost of holding reserves becomes $\lambda_j(f - r_{IOR})$.

2.5 Equilibrium

The Fed funds rates, f , is chosen exogenously by monetary policy makers. For a given Fed funds rate, f , each bank chooses its optimal deposit rate, r_j^* , and each depositor chooses its optimal investment, j^* , such that the deposit market clears. The lending market clears at equilibrium lending spreads, l_j^* .

2.6 Numerical example

Before I take the model to the data, it is useful to use a set of numerical examples to show how monetary policy is transmitted in a banking system with both commercial and shadow banks.

First, consider the case in which there is no friction in the banking system. In this benchmark case, banks have no market power, so the markups are zero. In addition, there is no reserve requirement. In this frictionless case, the deposit rates equal the bond market interest rates.

Imperfect competition creates a wedge between bond-market interest rates and deposit rates as banks exercise their market power. However, imperfect competition alone cannot generate the shadow banking channel in which shadow banks move in the opposite direction to commercial banks. To see this point, I solve the model assuming depositors are homogeneous. Table 6 presents the parameters. For simplicity, I assume that banks have the same cost structure and a constant lending spread.²⁸ The first row of Figure 5 shows the result. When depositors are homogeneous, the two banking sectors respond to monetary policy in a very similar way. Commercial banks on average charge a higher spread than shadow banks but the difference in spreads is stable over monetary cycles.

The second row of Figure 5 introduces depositor heterogeneity. In this case, banks exhibit different responses to monetary policy. When the Fed raises interest rates, commercial banks widen their spreads because cash becomes more expensive to hold for transaction-oriented depositors. However, shadow banks maintain tight spreads to keep their yield-oriented depositors from switching to bonds. This gives rise to the shadow banking channel documented in Section 1, in which shadow banks pass through more rate increases to depositors than commercial banks, resulting in an expansion of shadow bank deposits during monetary tightening cycles.

The difference in depositor clienteles is not the only institutional feature that could explain the different responses to monetary policy across shadow and commercial banks. An alternative channel is the reserve requirement as featured in the classical bank lending channel (Bernanke and Blinder 1988). The third row of Figure 5 shows the case in which shadow banks and commercial banks have different reserve requirements. As illustrated in Equation (8), higher Fed

²⁸ One way to rationalize the assumption of a constant lending spread is that loans have a constant convenience premium over bonds for firms, and firms demand for loans with infinite elasticity. In equilibrium, the loan spread equals the constant convenience premium and firms are indifferent between bonds and loans.

Table 6
Parameters for the numerical examples

	σ_α	σ_β	α	x_{cash}	x_{bond}	x_{cb}	x_{sb}	γ_1	γ_2	w_1	w_2
No heterogeneity	0.0	0.0	2.0	0.0	-2.5	-0.3	-1.8	0.0	1.0	f	1.0
Heterogeneous depositors	3.8	-2.4	2.0	0.0	-2.5	-0.3	-1.8	0.0	1.0	f	1.0
Reserve requirement	0.0	0.0	2.0	0.0	-2.5	-0.3	-1.8	0.1	1.0	f	1.0

This table presents the parameter values of the numerical examples from Figure 5. Each row presents the set of parameters for a different model.

funds rates increase the opportunity cost of holding reserves, $\lambda_j f$, which makes commercial bank deposits more costly to produce compared to shadow bank deposits.²⁹ Therefore, high Fed funds rates lead to a contraction of commercial banks and an expansion in shadow banks.

Disentangling the effect of clienteles from the effect of reserve requirements is difficult, because both channels generate similar qualitative predictions. The challenge lends itself to a structural estimation approach in which the quantitative contribution of each channel can be evaluated by comparing counterfactuals in which each channel is switched off. I will describe the estimation in the following section.

3. Structural Estimation

In this section, I take the model to the data. Here, the goal is to estimate the primitive structural parameters and pin down the exact mechanism of monetary transmission. This will set the stage for the ensuing counterfactual analysis.

3.1 Parametrization

I choose a set of product characteristics, x_j , based on the belief that they are important and recognizable to depositors' choice. Product characteristics include number of branches, number of employees per branch, checking-writing dummy, and TED spreads.³⁰ Egan, Hortaçsu, and Matvos (2017) use CDS spreads of banks as a proxy of risk. However, because bank-level CDS spreads are not widely available for small commercial banks or any of the MMFs, I use the TED spread to capture the risks of the banking system. I interact the TED spreads with the banking sector dummies because commercial

²⁹ An alternative way to describe the required reserve mechanism is through the quantity of reserves: when the Fed reduces the supply of reserves, bank deposits go down via the fractional reserve logic. Using either the quantity of reserves or the Fed fund rate to measure monetary policy is equivalent, because the Fed funds rate is the price of reserves, and the price and quantity are jointly determined in the equilibrium (Bernanke and Blinder 1992). Formally, in the model, the aggregate demand for reserves is given by $R_d(f) = \sum_j \lambda_j d_j(f)$. The aggregate supply of reserves, R_s , is determined by the Fed. In the equilibrium, the quantity and price of the reserves are jointly determined by equilibrium condition $R_d(f) = R_s$. Using either f or R_s describe the monetary policy is equivalent.

³⁰ Notice that non-transaction-related characteristics, such as brand awareness, also affect depositors' decisions. These characteristics are not central to the mechanism, because they affect the level of demand, rather than the slope or curvature. These characteristics are absorbed by the residuals of demand functions.

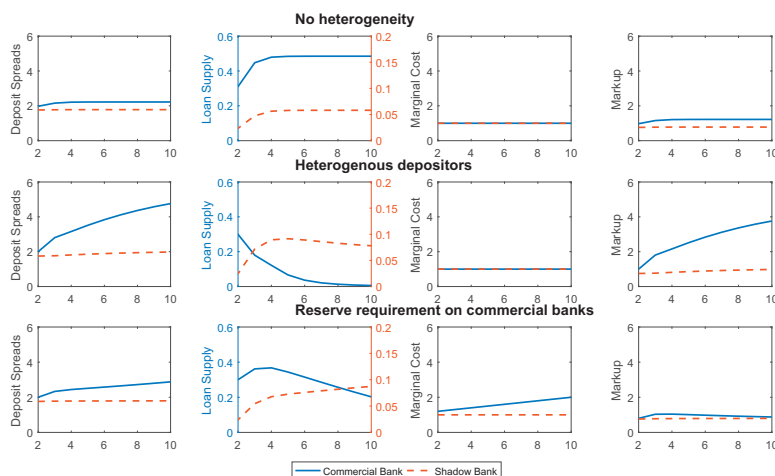


Figure 5

Numerical example: Deposit spreads and market shares

Each row is simulated using a different set of parameters from Table 6.

bank deposits are largely insured while shadow bank deposits are not. I also create a transaction dummy that equals one if a product is cash or commercial bank deposit and zero otherwise. I allow the sensitivity to yield, α_i , and the sensitivity to transaction dummy, β_i , to be heterogeneous across depositor types.³¹ Formally, I specify the type-specific sensitivity α_i and β_i as the sum of the mean sensitivity and a deviation term, $\alpha_i = \alpha + \sigma_\alpha v_i$, $\beta_i = \beta + \sigma_\beta v_i$. α and β are the means of yield sensitivity and transaction sensitivity. $\sigma = [\sigma_\alpha, \sigma_\beta]$ capture the dispersion among depositors. When σ_α and σ_β are zero, depositors are homogeneous. When σ_α and σ_β are larger, depositors become more heterogeneous. v_i is the type of depositors that follows a uniform distribution with a zero mean.

On the supply side, I specify the marginal cost as a linear function of cost shifters

$$c_j = \gamma' w_j + \omega_j \quad (10)$$

where w_j is a vector of observable supply shifters. The list of supply shifters for commercial banks includes salary expenses, expenses of fixed assets, and reserve costs. The reserve costs are defined as the product of the reserve ratio and the Fed funds rates as suggested by Equation (8). Previous literature also used these shifters (see, e.g., Dick 2008; Ho and Ishii 2011). The list of supply shifters for shadow banks include incurred management costs, other incurred operating costs, and the stand-alone dummy. These cost-shifters and their second-order polynomials also serve as instruments for the demand-side estimation.

³¹ In principle, I can allow all the sensitivity parameters to be heterogeneous. However, it would be infeasible to estimate all of them from the data.

To characterize the equilibrium in the deposit market, I need to know a set of preference parameters, α , β , and σ , which govern how depositors value different products. I also need to know a set of supply parameters, γ , which govern how much it costs to produce them. Formally, I can pin down these parameters by estimating the following two equations.

$$\delta_j(\sigma) = \alpha r_j + \beta' x_j + \xi_j \quad (11)$$

$$c_j = \gamma' w_j + \omega_j, \quad (12)$$

where $\delta_j = E[u_{i,j}]$ is the mean utility of product j across all depositors and ξ_j is an unobservable common demand shock to all depositors for product j .

I use the optimality conditions of depositors and banks to link unobserved utility and marginal costs to observable quantities, such as market shares and deposit rates. On the demand side, I numerically solve δ from a system of $J+1$ implicit equations implied by depositors' optimal choices using the fixed-point algorithm introduced by [Berry, Levinsohn, and Pakes \(1995\)](#) for a given value of σ

$$\underbrace{\delta_j(\sigma)}_{\text{Unobservable}} = d^{-1} \left(\underbrace{d_0}_{\text{Observable}} ; \underbrace{\sigma}_{\text{Unobservable}} \right)_j = \alpha r_j + \beta' x_j + \xi_j \quad (13)$$

where $d^{-1}(\cdot)$ is the inverse function of the demand equation 6, and d_0 is the vector of $J+1$ observable market shares.

On the supply side, I solve the unobservable marginal costs as the difference between deposit spreads and markups. The markups can be calculated once the preference parameters, α , σ , and β are estimated from the mean utility equation 11.

$$\underbrace{c_j}_{\text{Unobservable}} = p_j - \underbrace{\left(-\frac{\partial \log(d_j)}{\partial p_j} \right)^{-1}}_{\text{Unobservable}} \quad (14)$$

3.2 Identification

I first estimate the mean utility equation (11). Given the estimated demand-side parameters, I calculate the marginal costs and then estimate the cost coefficients of Equation (12).

A key challenge in identifying the demand parameters is that deposit rates are correlated with unobservable demand shocks, ξ_j . As a result, yield sensitivity α will be biased in an ordinary least squares (OLS) regression of mean utility, δ_j , on deposit rates, r_j . I follow the literature to use a set of cost shocks, z_j , as instrument variables for the deposit rates. The set of instruments includes salaries, expenses of fixed assets, incurred management costs, other incurred costs, and the stand-alone dummy for MMFs. The rationale is that these supply

shifters affect depositors' demand only through deposit rates instead of directly entering depositors' utility.

The moment condition of the mean utility equation is given by the orthogonality condition between the unobservable demand shocks, ξ_j , the product characteristics, x_j , and the cost shifters, z_j :

$$E[\xi_j[x_j, z_j]] = 0 \quad (15)$$

Formally, define θ as a vector of demand parameters, $\theta = [\sigma, \alpha, \beta]$, $Z = [x, z]$, W as a consistent estimate of $E[Z'\xi\xi'Z]$. The GMM estimator of the demand parameters is

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \xi(\theta)' Z W^{-1} Z \xi(\theta). \quad (16)$$

The algorithm first searches over the nonlinear parameter space of σ . Then, for a given σ , it solves $\delta_j(\sigma)$ through a fixed-point algorithm using the market share equation 13. Finally, the algorithm searches for a set of linear parameters α, β , which minimizes the GMM objective function. The three steps are repeated until the optimal set of parameters α, β , and σ is found.³²

Estimating the supply-side equation is more straightforward. The moment condition of the cost equation is given by the orthogonality condition between the idiosyncratic supply shock, ω_j , and observable cost shifters, w_j :

$$E[\omega_j w_j] = 0. \quad (17)$$

The supply parameters γ can be estimated by an OLS regression of the marginal cost on the supply shifters. Note that because the preference parameters are estimated from the first stage, the standard errors of the second stage are corrected using the approach in Newey and McFadden (1994).

3.3 Parameter estimates

Table 7 presents the demand parameters. Column 1 reports the logit model in which depositors are assumed to be homogeneous, and Column 2 reports the baseline BLP model where depositors are allowed to be heterogeneous. Later in Section 3.4, I will compare the predictions of the two models to show that depositor heterogeneity is crucial to explain the difference between shadow and commercial banks. For now, I focus on the parameter estimates. The estimation shows that depositors like higher deposit rates, more branches, more employees per branch, and the checking-writing services, but dislike higher default risks. The BLP model estimates in Column 2 also show statistically

³² Although one can see relatively easily how α and β are identified, the identification of σ is worthy of further elaboration. Intuitively, σ measures the dispersion of depositors' sensitivity to yields and transaction convenience. A greater dispersion means that different banks have very different demand elasticities. Therefore, if we observe that the same change in deposit rates leads to quite different changes in market share, that implies that depositors are quite heterogeneous. Table 4 in the Online Appendix shows that market shares of shadow banks are much more sensitive to rate changes compared to commercial banks in a simple reduced-form regression.

significant dispersions in the yield and transaction sensitivities. The two dimensions of heterogeneity are negatively correlated, $cov(\alpha_i, \beta_i) = \sigma_\alpha \sigma_\beta < 0$. This implies that depositors who have a higher sensitivity to transaction services tend to have a lower sensitivity to yields. Note that some estimates of the logit model are different from the BLP model. This is due to the homogeneous depositor assumption, which generates a time-varying bias in the mean utility. Specifically, when the Fed funds rates are high, yield-sensitive depositors leave the banking system and the yield-sensitivity of average depositor becomes lower. However, in the logit model, the yield-sensitivity is constant by assumption. This leads to an upward bias in the estimates of mean utility when the Fed funds rates are high. The opposite happens when the Fed funds rates are low.

Table 8 shows the summary statistics of the demand elasticity implied by the estimated demand parameters from the BLP model. The median own-rate elasticity of commercial banks is 0.757. The median own-rate elasticity of MMFs is 2.072, which is almost 3 times as large as that of commercial banks. This estimate suggests that in the equilibrium, yield-sensitive depositors endogenously sort into MMFs, while transaction-sensitive depositors endogenously sort into commercial banks.

Table 9 presents the estimated cost coefficients of the logit model (Column 1) and the BLP model (Column 2). The coefficients of reserve costs, salary expenses, expenses of fixed assets, incurred management costs, and other incurred operating costs are all positive and significant, consistent with our prior. In addition, stand-alone MMFs face higher costs than MMFs in a broader financial conglomerate. As discussed above, some estimates of the logit model differ from the BLP model because of the assumption of homogeneous depositors.

3.4 Transmission mechanism

3.4.1 Decomposing deposit spreads. Next, I use the estimated model to analyze the transmission channels of monetary policy. Figure 6 shows the deposit rates for commercial and shadow banks over time predicted by the BLP model. The model generates different pass-throughs from the Fed funds rates to deposit rates between commercial and shadow banks. The magnitude matches the data closely. Given that the parameters are primarily identified by the cross-section variations, it is remarkable that the model is able to match the different time-series variations for shadow and commercial banks.

As discussed in Section 2.6, the deposit rate pass-through is related to the pricing of deposits. The price of deposits, or the deposit spreads, can be decomposed to markups and marginal costs. The dashed line in Figure 7 shows the difference in marginal costs between commercial and shadow banks over time. Shadow banks on average incur 0.4% lower marginal costs than commercial banks, consistent with our intuition. Interestingly, the difference in marginal costs between shadow and commercial banks is quite stable over

Table 7
Demand parameter estimations

	Logit	BLP
Yield sensitivity (α)	1.264*** [0.050]	2.272*** [0.177]
Number of branches (β_1)	1.200*** [0.012]	1.169*** [0.010]
Number of employees per branch (β_2)	0.992*** [0.022]	0.979*** [0.021]
Check-writing	0.119*** [0.031]	0.230*** [0.047]
TED*CB (β_3)	-1.316* [0.725]	-0.406 [0.697]
TED*MMF (β_4)	-2.405*** [0.734]	-1.652** [0.706]
Yield sensitivity dispersion (σ_α)		4.236*** [0.463]
Transaction sensitivity dispersion (σ_β)		-3.282*** [0.961]
Time-Sector FE	Y	Y
Observations	12,674	12,674
Adj. R^2	.419	.848

This table presents the estimates of demand parameters of the structural model. Column 1 reports the logit model, and Column 2 reports the BLP model. The sample is a panel of U.S. commercial banks and MMFs from 1995 to 2012. Robust standard errors are reported in brackets and are clustered by time. * $p > .1$; ** $p > .05$; *** $p > .01$.

Table 8
Own-rate elasticity

	$\Delta \log(s_{cb})$	$\Delta \log(s_{sb})$
Δr_{cb}	0.732 (0.622)	
Δr_{sb}		2.082 (0.566)

This table presents the median and standard deviation (in brackets) of own-rate elasticity of commercial and shadow banks estimated from the BLP model. Each entry gives the percent change of the market share of a bank with a 1% change of its own deposit rates.

monetary cycles, implying that the opportunity cost of reserves incurred by commercial banks, λf , is quite small. This is not surprising because the effective reserve ratio, λ , is only 1.9% of the deposits as shown in Table 1. Given such a low reserve requirement, a 1% increase in the Fed funds rates translates into a 1.9 basis point increase in the reserve cost for commercial banks, which is too small to explain the variations in their deposit spreads. This result shows that the reserve mechanism in the classical bank lending channel cannot quantitatively explain the different monetary transmission for commercial and shadow banks, at least in recent years.

The solid line in Figure 7 shows the difference in markups between commercial and shadow banks. On average, shadow banks charge lower markups than commercial banks because the depositor clientele of shadow banks is more yield sensitive. The markups also significantly vary over time with different patterns across the two banking sectors. When the Fed raises the

Table 9
Supply parameter estimations

	(1) Logit	(2) BLP
	CB	
Expense of fixed assets (γ_1)	0.803*** [0.087]	0.484*** [0.055]
Salaries (γ_2)	0.238*** [0.029]	0.144*** [0.018]
Reserve cost (γ_3)	1.417*** [0.120]	0.904*** [0.075]
	MMF	
Incurred management costs (γ_4)	0.575*** [0.074]	0.421*** [0.046]
Incurred other operating costs (γ_5)	0.466*** [0.079]	0.352*** [0.049]
Standalone dummy (γ_6)	0.061*** [0.023]	0.033** [0.014]
Time FE	Y	Y
Sector FE	Y	Y
Adj. R^2	.651	.626
Observations	12,674	12,674

This table presents the estimates of supply parameters of the structural model. Column 1 reports the logit model, and Column 2 reports the BLP model. The sample is a panel of U.S. commercial banks and MMFs from 1995 to 2012. Robust standard errors are reported in brackets and are clustered by time. * $p > .1$; ** $p > .05$; *** $p > .01$.

Fed funds rates, commercial banks substantially increase their markups. This is consistent with the deposits channel proposed by [Drechsler, Savov, and Schnabl \(2017\)](#). In contrast, shadow banks keep charging a tight markup throughout monetary cycles. The differences in markups between the two banking sectors largely explain the difference in deposit spreads over monetary cycles.

To understand why two types of banks set different markups over monetary cycles, I shut down depositor heterogeneity by setting σ to zero. This setting corresponds to the logit model. The bottom panel of Figure 7 shows that the difference in the markups between the two banking sectors becomes almost zero. This result suggests that depositor heterogeneity is important to generate different monetary transmission across the two banking sectors.

3.4.2 Choice of depositors. In this section, I examine the choices of different types of depositors. I classify depositors with above-median yield sensitivity as yield-oriented depositors and depositors with below-median yield sensitivity as transaction-oriented depositors.³³

Figure 8 plots the probability of choosing commercial or shadow banks over time for each type of depositors. The first observation is that yield-sensitive depositors are on average more likely to choose shadow banks, while transaction-oriented depositors are more likely to choose commercial banks.

³³ Notice that the distribution of depositors' preferences is fixed over time in the baseline model. Later in the counterfactual section, I will explore what would happen if the composition of depositors were to change.

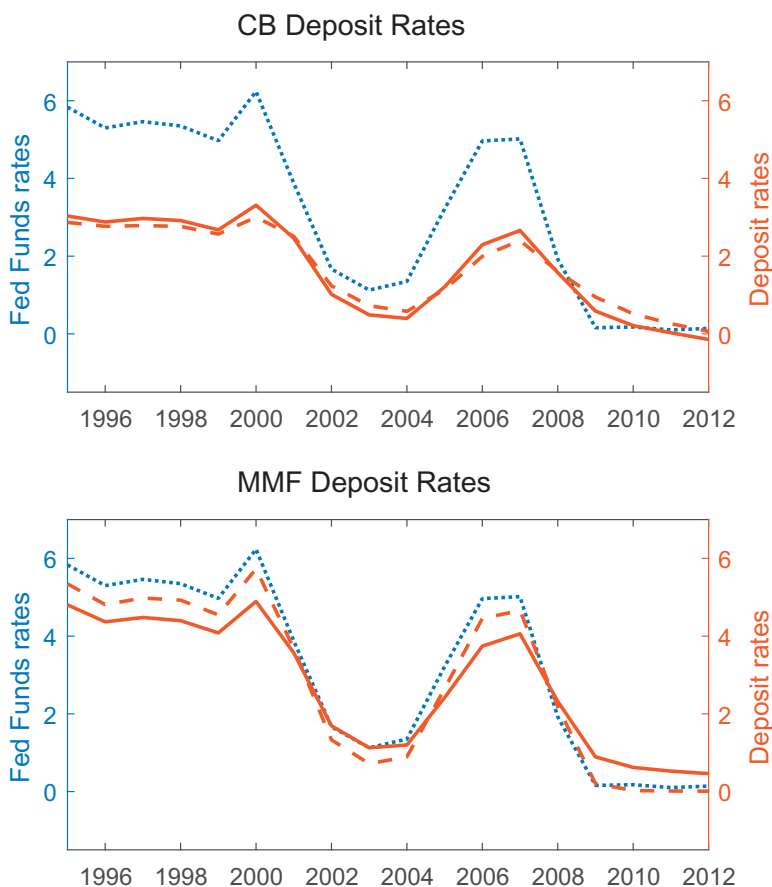


Figure 6
Model fit

This figure shows deposit rates of commercial and MMFs as predicted by the structural model and in the data. The model is estimated using institution-level data on U.S. commercial banks and MMFs from 1995 to 2012.

This is consistent with the finding in Table 8 that shadow banks face much more elastic demand than commercial banks. The second observation is that the choice probabilities vary substantially over time. When the Fed funds rates are low, yield-oriented depositors are more likely to choose commercial banks because both types of banks offer similar rates and commercial banks offer better transaction services. When the Fed funds rates are high, the spread between shadow and commercial bank deposit rates widens so the yield-oriented depositors switch to shadow banks. In contrast, transaction-oriented depositors stick to commercial banks all the time as their preference on transaction convenience dominates variations in deposit rates.

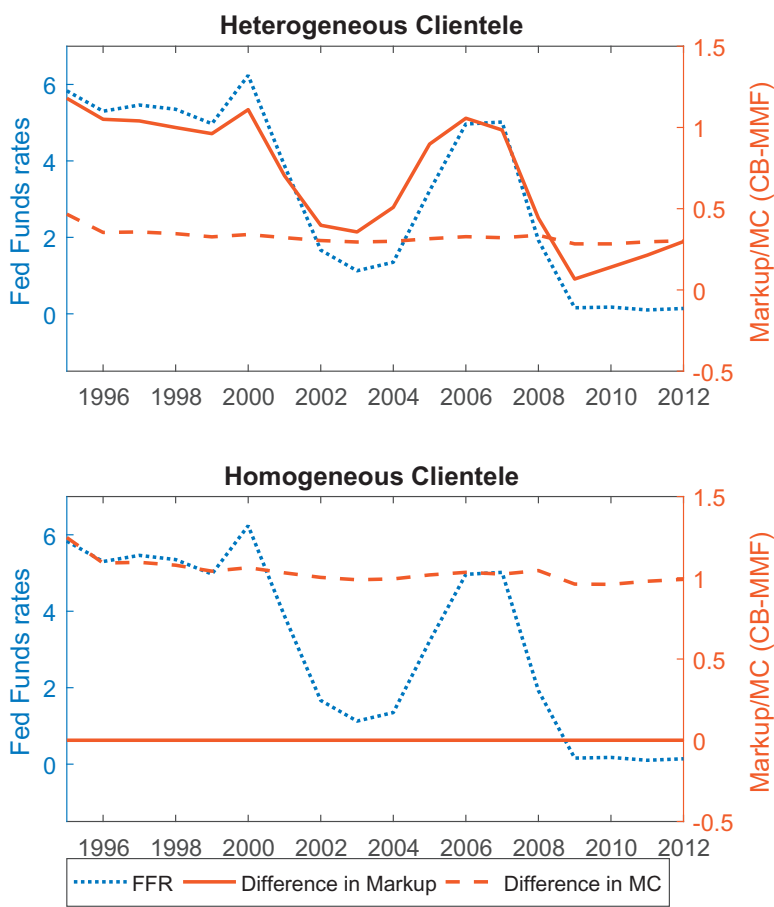


Figure 7
Difference in markups and marginal costs (CB-MMF)
This figure shows the difference in average markups and marginal costs between commercial and shadow banks estimated by the structural model. The model is estimated using institution-level data on U.S. commercial banks and MMFs from 1995 to 2012.

3.5 Alternative explanations

In the above analysis, I have shown that the difference in depositor clienteles between shadow and commercial banks can explain their different responses to monetary policy. However, many other institutional differences stretch across the banking sectors, and some of them may also explain these different responses. In this section, I examine some of the alternative explanations.

The first intuitive candidate is reserve requirements. As discussed in Section 2.6, commercial banks are subject to reserve requirements, but shadow banks are not. Monetary policy may have different impacts across banking sectors through the cost of holding reserves. The reserve-based explanation is unlikely

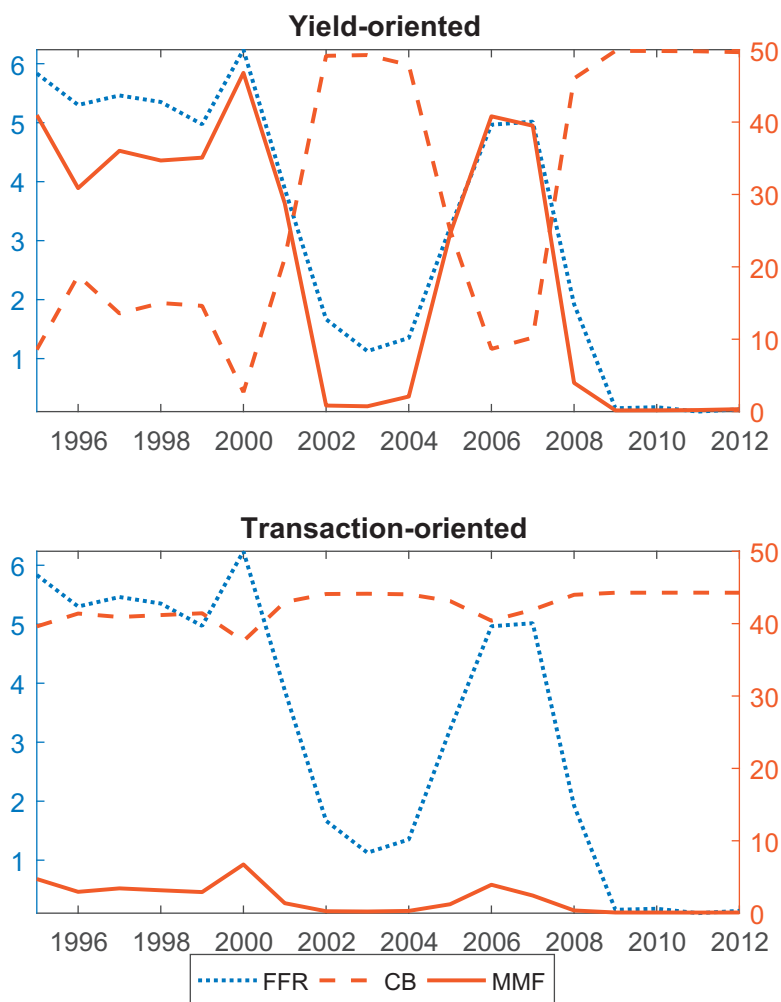


Figure 8

Choice probability of depositors by type

This figure shows the estimated probability for yield-oriented and transaction-oriented depositors to choose commercial banks or MMFs over time. The model is estimated using institution-level data on U.S. commercial banks and MMFs from 1995 to 2012.

to quantitatively explain the empirical finding. Technological innovations and regulatory reforms in the past three decades have rendered reserve requirements less important.³⁴ The structural model provides more concrete evidence supporting this view. In panel B of Figure 9, I shut down the reserve channel

³⁴ For example, sweep technology allows banks to easily transfer funds from transaction accounts to savings accounts to avoid reserve requirements (Teles and Zhou 2005).

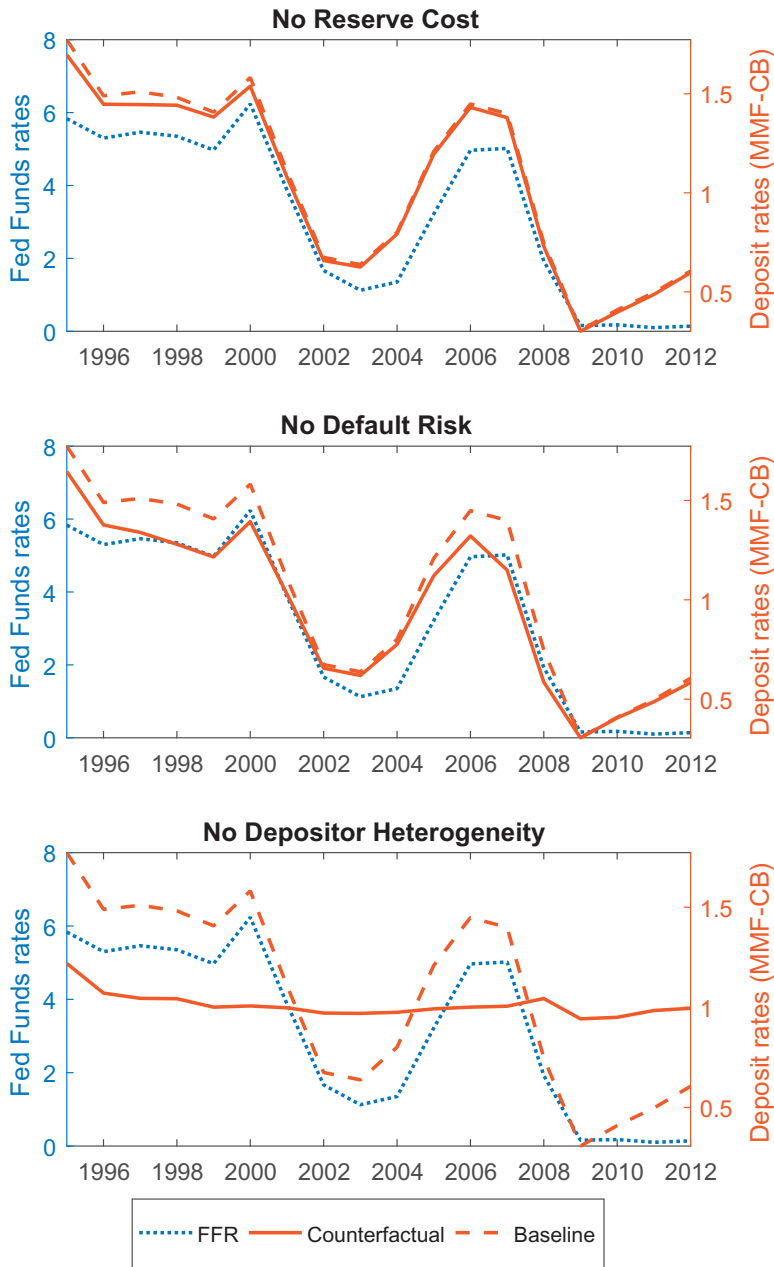


Figure 9
Decomposition of monetary transmission channels
This figure shows the difference in deposit rates between commercial and shadow banks in counterfactual simulations in which the bank reserve cost, default risk, or depositor heterogeneity is switched off, respectively.

by assuming the reserve requirement for commercial banks is zero. We can see that the procyclical pattern of the spread between shadow and commercial bank deposit rates hardly changes.

The second potential explanation for the different response to monetary policy by shadow banks is default risk. Shadow bank deposits are not insured by the FDIC. Therefore, in periods of crisis, depositors may transfer their money from shadow banks to commercial banks. Because the Fed usually cuts the Fed funds rates when the banking system is under distress, we may find a positive correlation between the Fed funds rates and deposit flows. To examine this alternative channel, in panel C of Figure 9, I eliminate the risk channel by setting the loading coefficients on TED spreads to zero. Comparing the simulated value with the real data, the relative deposit rates hardly change after shutting down this risk channel.

4. Counterfactuals

Having shown the transmission mechanism of monetary policy through shadow banks, I now conduct a set of counterfactual exercises to study the implications of changes in market structure for monetary transmission.

4.1 Shadow banks and effectiveness of monetary policy

There is a long-standing concern among policy-makers that financial innovation may undermine monetary control of the central bank. A classical paper on this issue, [Tobin and Brainard \(1963\)](#), argues that financial intermediaries that are not subject to direct monetary control of the central bank may dampen the effect of monetary policy. However, the mechanisms featured in [Tobin and Brainard \(1963\)](#) have become less relevant in the modern banking system: interest ceilings on commercial bank deposits were gradually phased out in the 1980s and the reserve requirement has become less binding since the 1990s. As shown in Section 3, monetary policy influences commercial banks largely through the deposits channel rather than the reserve requirement. In the modern banking system, in which interest ceilings and reserve requirements are not the defining difference between commercial and shadow banks anymore, how do shadow banks affect monetary policy transmission?

To answer this question, I simulate a counterfactual economy without shadow banks. I calculate the aggregate supply of bank loans and then calculate its sensitivity to the Fed funds rates. Row 2 of Table 10 reports the result. Comparing the counterfactual economy without shadow banks with the baseline economy with shadow banks in row 1, the presence of the shadow banking sector reduces the sensitivity of loan supply to the Fed funds rates by almost one-third.³⁵

³⁵ Note that this estimate does not mean that the overall impact of monetary policy is dampened by this amount because the calculation does not include neoclassical transmission channels through the bond market. It means that monetary transmission through the banking system is dampened by this amount.

How do shadow banks dampen the impact of monetary policy? In the counterfactual economy, yield-sensitive depositors do not have a buffer from shadow banks. When the Fed raises interest rates, yield-sensitive depositors flow out of the banking system altogether, leading to a reduction in the supply of loanable funds. In contrast, in an economy with shadow banks, yield-sensitive depositors can switch within the banking system from commercial banks to shadow banks so the total lending does not fall as much. In addition, the presence of shadow banks forces commercial banks to pay more competitive rates, which also dampens the impact of monetary policy by limiting the deposits channel.³⁶

This result relates to a strand of macroeconomic literature that studies the evolution of monetary transmission mechanisms over time. This literature documents that the effect of monetary policy on aggregate real activity seems to have become smaller in the post-1990s compared to the earlier period (Boivin, Kiley, and Mishkin 2011). Existing explanations include the changes of policy makers' focus and the changes in housing market credit conditions. My result suggests that the rise of the shadow banking sector also could be a contributing factor.

This result also relates to a strand of literature that studies how shadow banking affects the effectiveness of contractionary policies. Hachem and Song (2017) study the effect of bank liquidity regulation. They show that deposits migrate to the shadow sector as a result of stricter bank liquidity standards, so much so that the net effect of the tightening is an increase in total credit. Related to Hachem and Song (2017), this paper considers the transmission of some contractionary policies to loan supply through shadow banks. The difference is that this paper focuses on monetary policy, has a different mechanism based on heterogeneous clienteles, and is not claiming that the net effect on total credit is positive.

4.2 Increase in concentration

The concentration of the banking sector has substantially increased since the branching deregulation of the 1980s. Such increase in concentration may have important implications for monetary transmission.

To explore the effect of bank market concentration on monetary policy transmission, I simulate a set of counterfactual economies with various levels of concentration for the commercial and shadow banking sectors. Row 3 of Table 10 reports an economy in which all the commercial banks are merged into one monopoly bank, while the shadow banking sector maintains the same concentration as the real data. I find that this monopoly commercial bank charges substantially higher deposit spreads than the baseline case. The sensitivity of the aggregate loan supply to momentary policy also increases

³⁶ Note that such endogenous response is absent from the reduced-form regression in which we simply exclude the shadow bank deposits.

Table 10
Counterfactual simulation

	CB spreads	MMF spreads	$\frac{\Delta \log L}{\Delta FFR}$	Change wrt baseline
Baseline	1.230	0.419	-0.120	—
No MMFs	1.222	—	-0.154	27.95%
More concentrated CB	1.844	0.443	-0.197	63.63%
Less concentrated CB	1.202	0.418	-0.117	-2.49%
More concentrated MMF	1.229	0.445	-0.122	1.33%
Common owner	1.209	0.427	-0.131	8.98%
More regulated CB	1.554	0.899	-0.188	56.34%
More yield-sensitive depositors	1.066	0.301	-0.129	7.17%
More yield-insensitive depositors	1.269	0.607	-0.111	-7.25%

This table presents the result of counterfactual simulations. Each row represents a counterfactual scenario. Column 1 and 2 reports the average deposit spreads for commercial and shadow banks. Column 3 reports the sensitivity of loan supply to changes in the Fed funds rates. Column 4 reports the percentage change in the loan-FFR sensitivity with respect to the baseline simulation in row 1. The parameters are estimated from a panel of U.S. commercial banks and MMFs from 1995 to 2012.

substantially. This is because the deposits channel becomes stronger when commercial banks have more market power.

Row 4 of Table 10 reports an economy in which the commercial banking sector becomes less concentrated. Specifically, the counterfactual economy is populated with 100 small commercial banks with the same size. The HHI of the commercial banking sector drops from 667 in the baseline case to 100. In this counterfactual economy, commercial banks charge lower deposit spreads on average, and the sensitivity of the aggregate loan supply to monetary policy becomes lower.

Row 5 of Table 10 reports an economy in which all the shadow banks are merged into one monopoly shadow bank. The monopoly shadow bank charges higher deposit spreads than the baseline case. Monetary transmission is enhanced because the monopoly shadow bank does not compete as fiercely as before. Interestingly, the deposit spreads of the monopoly shadow banks are lower than the monopoly commercial bank in the previous case in row 4. The reason is that shadow banks face the yield-sensitive clientele who are ready to switch to bonds. Such clientele prevents the monopoly shadow bank from exercising their market power. This result shows that the difference in market power between shadow and commercial banks largely comes from the difference in clienteles. This clientele effect is not captured by the commonly used reduced-form measure of market power, HHI, which only captures the effect of concentration. In contrast, the structural IO framework captures both effects.

Row 6 of Table 10 reports an economy in which shadow and commercial banks are jointly owned by holding companies. Specifically, I assume that there are 15 holding companies. Each holding company owns one commercial bank and seven MMFs to match the concentration of the baseline case. The key difference from the baseline case is that the rate-setting decisions are made at the holding company level rather than the subsidiary bank or fund level. I find that the average deposit spreads of shadow banks increase because their

pricing of deposits internalize the cannibalization effects to the commercial bank subsidiary within the same holding company. The monetary transmission is enhanced compared to the baseline because shadow banks will not compete as fiercely to undercut commercial banks.

4.3 Tighter liquidity regulation on commercial banks

After the 2007–2009 financial crisis, banking regulators around the world imposed tighter liquidity regulation on commercial banks. For instance, the Basel Committee proposed the Liquidity Coverage Ratio regulation (LCR), which requires banks to hold sufficient amount of high-quality liquid assets (HQLA) to deal with adverse liquidity shocks. This requirement lowers the asset returns of commercial banks because the high-quality liquid assets usually have lower yields. Here, I use the structural model to study the effect of tighter liquidity regulation on the transmission of monetary policy.

Row 7 of Table 10 reports an economy in which commercial banks face tighter liquidity regulation, which reduces the asset returns of commercial banks by 0.3%.³⁷ I find that the deposit spreads of commercial banks increase compared to the baseline because the lending opportunity is not as attractive as before. Higher regulatory costs drive the marginal depositors of commercial banks to shadow banks. Because these marginal depositors are less yield-sensitive than the original shadow bank clientele, the demand for shadow bank deposits also becomes more inelastic and deposit spreads of shadow banks increase as a result. The sensitivity of aggregate loan supply to the Fed funds rates becomes higher because deposit spreads vary more over monetary cycles.

4.4 Shift in composition of depositors

Cash holdings of U.S. companies have grown enormously over the past 40 years. Because corporate treasurers are usually more yield sensitive than mom-and-pop depositors, the increase in corporate cash holding implies an increase in the fraction of yield-sensitive depositors. How does such shift in depositor composition affect the transmission of monetary policy?

Row 8 of Table 10 reports an economy in which the fraction of yield-sensitive depositors increases by 25%. There are two countervailing forces. On the one hand, because depositors become more sensitive to yields on average, deposit quantities will change by a larger amount in response to the same change in deposit spreads. On the other hand, because banks are able to exercise less market power when faced with more yield-sensitive depositors, the changes in deposit spreads over monetary cycles will be smaller. Overall, the first effect dominates and the sensitivity of aggregate loan supply to the Fed funds rates increases as the fraction of yield-sensitive depositors increases. Row 9 of Table

³⁷ Specifically, *Sundaresan and Xiao (2019)* show that the fraction of the high-quality liquid assets after the crisis has increased by 15% compared to the pre-crisis level. The average loan spread is around 2%. This gives a change in the asset returns of 0.3%.

10 reports the opposite case in which the fraction of yield-insensitive depositors decreases by 25%. Deposit spreads increase and the aggregate loan supply becomes less sensitive to the Fed funds rates.

5. Policy Implications

5.1 Monetary policy as a macroprudential policy tool

Prior to the 2008–2009 financial crisis, the consensus among policy makers was that monetary authority should focus on the dual mandate of price stability and maximum employment (Smets 2013). However, this consensus has been challenged by an alternative view that took shape after the financial crisis, which argues that monetary policy also should be used as a macroprudential tool to promote financial stability (Borio and Zhu 2012; Stein 2012; Ajello et al. 2015). Proponents of this view contend that by tightening monetary policy, the central bank can curb, among other things, the creation of money-like liabilities by the banking system. The unique advantage of monetary policy over banking supervision or regulation is that monetary policy can “get in all of the cracks” because all the financial institutions face the same set of market interest rates no matter whether they are regulated or not (Stein 2013). On the other hand, the potential drawbacks of using monetary policy as a macroprudential tool is also discussed (Stein 2012; Yellen 2014). Stein (2012) shows that the lack of reserve requirement on shadow banks may limit the effectiveness of monetary policy as a macroprudential tool. Yellen (2014) argues that monetary policy has significant limitations as a tool to promote financial stability because its effects on financial vulnerabilities are not well understood and are less direct than a regulatory or supervisory approach.

My finding contributes to this debate in two aspects. First, it shows that monetary policy may not be an effective tool for influencing the creation of money-like assets in the presence of the shadow banking sector, because shadow banks partially offset the tightening effect on commercial banks. Second, monetary tightening may actually increase systemic risk because shadow banks are not protected by deposit insurance.³⁸ Therefore, my paper supports the view that “monetary policy is too blunt a tool to address possible financial imbalances” as argued by Bernanke (2011) and Yellen (2014).

5.2 Impact of shadow banks on depositor surplus

Shadow banks are often viewed as a way of regulatory arbitrage (Acharya, Schnabl, and Suarez 2013; Plantin 2014). An implication of this view is that regulators should limit shadow banks because they do not serve new economic functions that commercial banks cannot serve. Recent regulatory reforms seem

³⁸ My model only isolates the effect of monetary policy on the composition of commercial and shadow bank deposits. It does not quantify the effect of such composition changes on the probability of a run on the financial system.

to reflect this view. For instance, the 2016 Money Market Fund Reform prohibits prime MMFs from providing deposit-like features, such as fixed net asset values. This regulation leads to large deposit outflows from the affected MMFs.

What is missing from this narrative is that shadow banks may create economic values if the competition in the traditional commercial banking sector is far from perfect. To see this point, I compare the actual supply of commercial bank deposits with a perfectly competitive economy in which deposit spreads are set at the marginal costs. I find that the actual supply accounts for only 21% of counterfactual amount under perfect competition. This result shows that the competition in the actual economy is far from perfect and commercial banks substantially restrict their supply to earn monopoly rents. In this case, the entry of shadow banks may increase depositor surplus by pushing the equilibrium quantity closer to the perfectly competitive case.³⁹

To assess the impact of shadow banking on depositor surplus, I compare the depositor surplus in a counterfactual economy without shadow banks with the baseline case. Specifically, I follow [Nevo \(2001\)](#) to compute the expected utility for each type of depositor i from its optimal choice.

$$E \left[\max_{j \in \{0, 1, \dots, J\}} u_{i,j} \right] = \ln \left(\sum_{j=0}^J \exp(\alpha_i r_j + \beta'_i x_j + \xi_j) \right) \quad (18)$$

Then, I divide expected utility by the yield sensitivity to calculate the equivalent utility in the unit of deposit rates. Finally, I aggregate across depositor types to calculate the total depositor surplus.

$$\text{Depositor Surplus}_t = \sum_i \mu_i \frac{1}{\alpha_i} E \left[\max_{j \in \{0, 1, \dots, J\}} u_{i,j} \right] \quad (19)$$

I compare the surplus in the counterfactual economy with the actual economy. The entry of shadow banks on average generates 2.86 basis points per dollar per year in the sample period. I also find that the increase in depositor surplus is larger when the Fed funds rates are high, which is consistent with the previous result that commercial banks enjoy greater market power during these periods.⁴⁰ Note that the above calculation only considers depositor surplus. It does not account for banks' profits and potential externalities to the rest of the economy. Therefore, the question of whether shadow banks increase overall social welfare remains unanswered. Nevertheless, shadow banks do indeed seem to create some benefits that should be considered during policy deliberation.

³⁹ In general, the entry of new competitors can significantly increase the consumer surplus in a concentrated industry even if the new entrants offer the same or even inferior products compared with incumbents. In this model, I do not endogenize the entry decision. But one can imagine that in a more general model with endogenous entry, high operating and regulatory costs create a barrier for new commercial banks to enter, resulting in insufficient competition. In contrast, the entry costs for shadow banks are likely to be relatively lower. Therefore, shadow banks emerge in the equilibrium. Modeling and quantifying the entry cost, however, is beyond the scope of this paper.

⁴⁰ [Online Appendix](#) Figure 7 shows the time-series plot of the depositor surplus due to shadow banks.

6. Conclusion

This paper documents a new monetary transmission mechanism: the shadow banking channel. I find that money supply from shadow banks expands when the Fed raises interest rates. This is at odds with the conventional wisdom in the commercial banking sector that monetary tightening reduces money creation. I show that this new channel is a result of deposit competition between commercial and shadow banks in a market with heterogeneous depositors. Fitting my model to institution-level commercial bank and MMF data shows that this channel dampens the impact of monetary policy on money supply. I also explore the implications of shadow banking for macroprudential policy. My results raise the possibility that using monetary tightening as a macroprudential tool could unintentionally increase the fragility of the banking system, because monetary tightening drives deposits from the insured commercial banking sector to the uninsured shadow banking sector.

This paper highlights the importance of industrial organization of the banking system. Shadow banks provide a valuable alternative to commercial bank deposits that, due to their market power, pay too little to depositors. In this sense, shadow banks may create economic value by making the banking system more competitive.

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